



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

**NDIC - DATEX II Measurement Site
Table Publication - Predefined
Equipment Set**

Release 1.0rev0

Národní Dopravní Informační Centrum

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Jan Vlčinský, Petr Bureš

The specification defines format for publishing predefined equipment data used to reference status oriented events (detector data, weather data etc.).

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [\[d2ref\]](#). For additional information see [\[d2supp\]](#).

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

Described format allows publishing of predefined equipment data used to reference status oriented events (detector data, weather data etc.).

2.1 Publication type MeasurementSiteTablePublication

Format is based on DATEX II publication of type *MeasurementSiteTablePublication* [16157-5]. The profile uses most of data structures, provided by this type of publication and extends it by proprietary location referencing.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsd=
  ↳ "http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
  ↳ instance">
3   <exchange>
4     <supplierIdentification>
5       <country>cz</country>
6       <nationalIdentifier>DRU</nationalIdentifier>
7     </supplierIdentification>
8   </exchange>
9   <payloadPublication lang="cs" xsi:type="MeasurementSiteTablePublication">
10    <feedType>cz-ndic_d2-measutrement-site-table-v1.0</feedType>
11    <publicationTime>2018-01-29T08:00:29+02:00</publicationTime>
12    <publicationCreator>
13      <country>cz</country>
14      <nationalIdentifier>DRU</nationalIdentifier>
15    </publicationCreator>
16    <headerInformation>
17      <confidentiality>noRestriction</confidentiality>
18      <informationStatus>real</informationStatus>
19    </headerInformation>
20    <!-- detector measurement table begin -->
```

2.2 Predefined equipment description (measurementSiteTable and measurementSiteRecord)

MeasurementSiteTablePublication requires description by means of class *measurementSiteTable*, which aggregates instances of class *measurementSiteRecord*. *measurementSiteTable* also name of the respective table.

Element *measurementSiteTable* describes a particular type of sites (usually), more particular types of sites or more infrastructure operators means more *measurementSiteTable* elements. Since the equipment described may change over time, the element is versioned and has unique identifier. Each *measurementSiteTable* element contains one or more *measurementSiteRecord* elements.

Textual description of each *measurementSiteRecord* could be located within a *measurementSiteName* element.

Each *measurementSiteRecord* element is describing different measuring equipment with different measuring capabilities *measurementSpecificCharacteristics* i.e. measuring of traffic flow, speed, density, etc. The capabilities of *measurementSiteRecord* element might change in time, so the element is versioned and has an identifier unique within the scope of one measurement table.

The capabilities *measurementSpecificCharacteristics* of the measuring equipment in particular includes:

- humidity,
- individual vehicle measurements (headway, speed, etc.),
- pollution,
- precipitation,
- pressure,
- radiation,
- road surface conditions,
- temperature,
- traffic concentration,
- traffic flow,
- traffic headway,
- traffic speed,
- traffic status,
- travel time,
- visibility and
- wind.

Some *measurementSpecificCharacteristics*, where it matters can be further specified by positional arguments and vehicle specification (i.e. traffic flow of heavy vehicles, road temperature in second lane). Each capability is indexed (uniquely within one equipment). The index is used for capability referencing from the dynamic publication.

```

21 <measurementSiteTable id="CZ_DRU_MST_DET" version="1">
22   <measurementSiteTableIdentification>Tabulka detektoru v HDRU</
↪ measurementSiteTableIdentification>
23   <measurementSiteRecord id="1307" version="1">
24     <measurementSiteName>
25       <values>
26         <value lang="cs">název umístění detektoru</value>
27       </values>
28     </measurementSiteName>
29     <!--start of overall characteristics: flow, occupancy and speed -->
30     <measurementSpecificCharacteristics index="1">
31       <measurementSpecificCharacteristics>
32         <specificMeasurementValueType>trafficFlow</specificMeasurementValueType>
33         <specificVehicleCharacteristics>
34           <vehicleType>anyVehicle</vehicleType>
35         </specificVehicleCharacteristics>
36       </measurementSpecificCharacteristics>
37     </measurementSpecificCharacteristics>
38     <measurementSpecificCharacteristics index="2">
39       <measurementSpecificCharacteristics>
40         <specificMeasurementValueType>trafficConcentration</
↪ specificMeasurementValueType>

```

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```

41      <specificVehicleCharacteristics>
42        <vehicleType>anyVehicle</vehicleType>
43      </specificVehicleCharacteristics>
44    </measurementSpecificCharacteristics>
45  </measurementSpecificCharacteristics>
46  <measurementSpecificCharacteristics index="3">
47    <measurementSpecificCharacteristics>
48      <specificMeasurementValueType>trafficSpeed</specificMeasurementValueType>
49      <specificVehicleCharacteristics>
50        <vehicleType>anyVehicle</vehicleType>
51      </specificVehicleCharacteristics>
52    </measurementSpecificCharacteristics>
53  </measurementSpecificCharacteristics>

```

2.3 Equipment location description (point or linear segment)

Each equipment location is specified within the element *measurementSiteLocation*.

Location is a point or linear segment on road network. Location type area is not used. The locations may be described by means of several methods, locations are described in line (not referenced to another publication)

```

213  <measurementSiteLocation xsi:type="Point">
214    <pointWithinLinearGNElement>
215      <sectionId>822309</sectionId>
216      <directionRelativeOnLinearSection>aligned</
↪ directionRelativeOnLinearSection>
217      <networkVersion>
218        <values>
219          <value lang="cs">16.12</value>
220        </values>
221      </networkVersion>
222      <distanceAlongLinearElement xsi:type="PercentageDistanceAlongLinearElement
↪ ">
223        <percentageDistanceAlong>0.662356475026882</percentageDistanceAlong>
224      </distanceAlongLinearElement>
225    </pointWithinLinearGNElement>
226  </measurementSiteLocation>

```

The position description methods are further described in the chapter "Location description".

TIME VALIDITY

The publication start of validity is defined by *publicationTime* for all predefined equipment in the publication. No other time record is contained in the publication. In case of new publication due to changes to predefined equipment set, existing equipment remains the same, new equipment get new id and updated equipment got its version increased.

11

```
<publicationTime>2018-01-29T08:00:29+02:00</publicationTime>
```

LOCATION REFERENCING

All equipment predefined in this publication has its location set up within element *measurementSiteLocation* with type *Point* or *Linear* that contains description of the equipment physical location in one or more location referencing methods.

```
213     <measurementSiteLocation xsi:type="Point">
214       <pointWithinLinearGNElement>
215         <sectionId>822309</sectionId>
216         <directionRelativeOnLinearSection>aligned</
↪ directionRelativeOnLinearSection>
217         <networkVersion>
218           <values>
219             <value lang="cs">16.12</value>
220           </values>
221         </networkVersion>
222         <distanceAlongLinearElement xsi:type="PercentageDistanceAlongLinearElement
↪ ">
223           <percentageDistanceAlong>0.662356475026882</percentageDistanceAlong>
224         </distanceAlongLinearElement>
225       </pointWithinLinearGNElement>
226     </measurementSiteLocation>
```

Three location description methods for a linear location and four location description methods for point location can be used to describe the location. The method use depends on method availability at the location. At least one location description should be used.

APPENDIX A: PUBLICATION EXAMPLE

5.1 measurement_site_table.xml - predefined location set with two locations

Publication contains two predefined equipments (in two tables, one for detectors and other for wheather stations).

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsd=
  ↪ "http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
  ↪ instance">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="MeasurementSiteTablePublication">
    <feedType>cz-ndic_d2-measutrement-site-table-v1.0</feedType>
    <publicationTime>2018-01-29T08:00:29+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </publicationCreator>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <!-- detector measurement table begin -->
    <measurementSiteTable id="CZ_DRU_MST_DET" version="1">
      <measurementSiteTableIdentification>Tabulka detektoru v HDRU</
  ↪ measurementSiteTableIdentification>
      <measurementSiteRecord id="1307" version="1">
        <measurementSiteName>
          <values>
            <value lang="cs">název umístění detektoru</value>
          </values>
        </measurementSiteName>
        <!--start of overall characteristics: flow, occupancy and speed -->
        <measurementSpecificCharacteristics index="1">
          <measurementSpecificCharacteristics>
            <specificMeasurementValueType>trafficFlow</specificMeasurementValueType>
            <specificVehicleCharacteristics>
              <vehicleType>anyVehicle</vehicleType>
            </specificVehicleCharacteristics>
          </measurementSpecificCharacteristics>
        </measurementSpecificCharacteristics>
      </measurementSiteRecord>
    </measurementSiteTable>
  </payloadPublication>
</d2LogicalModel>
```

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```

        </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="2">
        <measurementSpecificCharacteristics>
            <specificMeasurementValueType>trafficConcentration</
↪specificMeasurementValueType>
            <specificVehicleCharacteristics>
                <vehicleType>anyVehicle</vehicleType>
            </specificVehicleCharacteristics>
        </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="3">
        <measurementSpecificCharacteristics>
            <specificMeasurementValueType>trafficSpeed</specificMeasurementValueType>
            <specificVehicleCharacteristics>
                <vehicleType>anyVehicle</vehicleType>
            </specificVehicleCharacteristics>
        </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <!--end of overall characteristics: flow, occupancy and speed -->
    <!--start of vehicle flow per hour, classes: 16-21
    just wild guessing
    class 16 = car
    class 17 = lorry
    class 18 = trailer
    class 19 = van
    class 20 = bus
    class 21 = motorcycle
-->
    <measurementSpecificCharacteristics index="4">
        <measurementSpecificCharacteristics>
            <specificMeasurementValueType>trafficFlow</specificMeasurementValueType>
            <specificVehicleCharacteristics>
                <vehicleType>car</vehicleType>
            </specificVehicleCharacteristics>
        </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="5">
        <measurementSpecificCharacteristics>
            <specificMeasurementValueType>trafficFlow</specificMeasurementValueType>
            <specificVehicleCharacteristics>
                <vehicleType>lorry</vehicleType>
            </specificVehicleCharacteristics>
        </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="6">
        <measurementSpecificCharacteristics>
            <specificMeasurementValueType>trafficFlow</specificMeasurementValueType>
            <specificVehicleCharacteristics>
                <vehicleType>trailer</vehicleType>
            </specificVehicleCharacteristics>
        </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="7">
        <measurementSpecificCharacteristics>
            <specificMeasurementValueType>trafficFlow</specificMeasurementValueType>

```

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```

        <specificVehicleCharacteristics>
          <vehicleType>van</vehicleType>
        </specificVehicleCharacteristics>
      </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics index="8">
      <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficFlow</specificMeasurementValueType>
        <specificVehicleCharacteristics>
          <vehicleType>bus</vehicleType>
        </specificVehicleCharacteristics>
      </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="9">
      <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficFlow</specificMeasurementValueType>
        <specificVehicleCharacteristics>
          <vehicleType>motorcycle</vehicleType>
        </specificVehicleCharacteristics>
      </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <!--end of vehicle flow per hour, classes: 16-21 -->
    <!--start of occupancy, classes: 16-21 -->
    <measurementSpecificCharacteristics index="10">
      <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficConcentration</
    <specificMeasurementValueType>
        <specificVehicleCharacteristics>
          <vehicleType>car</vehicleType>
        </specificVehicleCharacteristics>
      </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="11">
      <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficConcentration</
    <specificMeasurementValueType>
        <specificVehicleCharacteristics>
          <vehicleType>lorry</vehicleType>
        </specificVehicleCharacteristics>
      </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="12">
      <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficConcentration</
    <specificMeasurementValueType>
        <specificVehicleCharacteristics>
          <vehicleType>trailer</vehicleType>
        </specificVehicleCharacteristics>
      </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <measurementSpecificCharacteristics index="13">
      <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficConcentration</
    <specificMeasurementValueType>
        <specificVehicleCharacteristics>
          <vehicleType>van</vehicleType>

```

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```

        </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<measurementSpecificCharacteristics index="14">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficConcentration</
↪specificMeasurementValueType>
        <specificVehicleCharacteristics>
            <vehicleType>bus</vehicleType>
        </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<measurementSpecificCharacteristics index="15">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficConcentration</
↪specificMeasurementValueType>
        <specificVehicleCharacteristics>
            <vehicleType>motorcycle</vehicleType>
        </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<!--end of occupancy, classes: 16-21 -->
<!--start average speed, classes: 16,18,19 -->
<measurementSpecificCharacteristics index="16">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficSpeed</specificMeasurementValueType>
        <specificVehicleCharacteristics>
            <vehicleType>car</vehicleType>
        </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<measurementSpecificCharacteristics index="17">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficSpeed</specificMeasurementValueType>
        <specificVehicleCharacteristics>
            <vehicleType>lorry</vehicleType>
        </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<measurementSpecificCharacteristics index="18">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficSpeed</specificMeasurementValueType>
        <specificVehicleCharacteristics>
            <vehicleType>trailer</vehicleType>
        </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<measurementSpecificCharacteristics index="19">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>trafficSpeed</specificMeasurementValueType>
        <specificVehicleCharacteristics>
            <vehicleType>van</vehicleType>
        </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<measurementSpecificCharacteristics index="20">

```

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```

    <measurementSpecificCharacteristics>
      <specificMeasurementValueType>trafficSpeed</specificMeasurementValueType>
      <specificVehicleCharacteristics>
        <vehicleType>bus</vehicleType>
      </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
  </measurementSpecificCharacteristics>
  <measurementSpecificCharacteristics index="21">
    <measurementSpecificCharacteristics>
      <specificMeasurementValueType>trafficSpeed</specificMeasurementValueType>
      <specificVehicleCharacteristics>
        <vehicleType>motorcycle</vehicleType>
      </specificVehicleCharacteristics>
    </measurementSpecificCharacteristics>
  </measurementSpecificCharacteristics>
  <!-- location description start -->
  <measurementSiteLocation xsi:type="Point">
    <pointWithinLinearGNElement>
      <sectionId>822309</sectionId>
      <directionRelativeOnLinearSection>aligned</
    directionRelativeOnLinearSection>
      <networkVersion>
        <values>
          <value lang="cs">16.12</value>
        </values>
      </networkVersion>
      <distanceAlongLinearElement xsi:type="PercentageDistanceAlongLinearElement
    ">
      <percentageDistanceAlong>0.662356475026882</percentageDistanceAlong>
    </distanceAlongLinearElement>
    </pointWithinLinearGNElement>
  </measurementSiteLocation>
  <!-- location description end -->
</measurementSiteRecord>
<!-- other measurement site records will be the same -->
</measurementSiteTable>
<!-- detector measurement table end -->
<!-- weather station measurement table begin -->
<measurementSiteTable id="CZ_DRU_MST_WEA" version="1">
  <measurementSiteRecord id="12" version="1">
    <measurementSiteName>
      <values>
        <value lang="cs">název umístění meteostanice</value>
      </values>
    </measurementSiteName>
    <!--precip_type + precipitation -->
    <measurementSpecificCharacteristics index="1">
      <measurementSpecificCharacteristics>
        <specificMeasurementValueType>precipitationInformation</
    specificMeasurementValueType>
      </measurementSpecificCharacteristics>
    </measurementSpecificCharacteristics>
    <!--road_status -->
    <measurementSpecificCharacteristics index="2">
      <measurementSpecificCharacteristics>
        <specificMeasurementValueType>roadSurfaceConditionInformation</

```

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```

↪specificMeasurementValueType>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<!--road_temp 1 + freez_temp -->
<measurementSpecificCharacteristics index="3">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>roadSurfaceConditionInformation</
↪specificMeasurementValueType>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<!--road_temp 2 + freez_temp -->
<measurementSpecificCharacteristics index="4">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>roadSurfaceConditionInformation</
↪specificMeasurementValueType>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<!--air_temp + dewpoint -->
<measurementSpecificCharacteristics index="5">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>temperatureInformation</
↪specificMeasurementValueType>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<!--humidity -->
<measurementSpecificCharacteristics index="6">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>humidityInformation</
↪specificMeasurementValueType>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<!--wind_dir + wind_sp-->
<measurementSpecificCharacteristics index="7">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>windInformation</
↪specificMeasurementValueType>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<!--visibility-->
<measurementSpecificCharacteristics index="8">
    <measurementSpecificCharacteristics>
        <specificMeasurementValueType>visibilityInformation</
↪specificMeasurementValueType>
    </measurementSpecificCharacteristics>
</measurementSpecificCharacteristics>
<!-- location description start -->
<measurementSiteLocation xsi:type="Point">
    <pointWithinLinearGNElement>
        <sectionId>822309</sectionId>
        <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
    <networkVersion>
        <values>
            <value lang="cs">16.12</value>
        </values>
    </networkVersion>

```

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```

    <distanceAlongLinearElement xsi:type="PercentageDistanceAlongLinearElement
    <percentageDistanceAlong>0.662356475026882</percentageDistanceAlong>
  </distanceAlongLinearElement>
</pointWithinLinearGNElement>
</measurementSiteLocation>
<!-- location description end -->
</measurementSiteRecord>
</measurementSiteTable>
<!-- weather station measurement table end -->
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *measurementSiteTable*, (in our case there are two) containing elements *measurementSiteRecord*.

Attribute *xsi:type="MeasurementSiteTablePublication"* in element *payloadPublication* determines, that this envelope is of type predefined equipment publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* states publication time of this publication as a whole.

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of predefined equipment

Predefined equipment is described in measurement site tables uniquely identified by id and version in the example *id="CZ_DRU_MST_DET"* for a detector table and *id="CZ_DRU_MST_WEA"* for weather station table.

Each table, in the example, contain just one element describing respective equipment capabilities. The identification of the equipment is unique within the table. Detector having *id="1307"* and *id="12"* for weather station.

Equipment description (CZ_DRU_MST_DET, 1307)

Element *measurementSiteRecord* contains a textual description of the equipment.

The equipment (traffic detector) has 21 capabilities defined. They are vehicle categorized versions of three traffic variables:

- traffic flow,
- traffic concentration,
- traffic speed.

First 3 capabilities *measurementSpecificCharacteristics* with indexes 1-3 describe flow, occupancy and speed for all vehicles. Remaining capabilities describe these traffic values categorized by vehicle type.

E.g. capabilities with indexes 4-9 describe traffic flow for:

- car,
- lorry,
- trailer,

- van,
- bus,
- motorcycle.

The capabilities *measurementSpecificCharacteristics* with indexes 10-15 describe categorized occupancy and 16-21 categorized speed.

Element *measurementSiteLocation* of type *xsi:type="Point"* contains individual location description methods, in our case just one the description using Global Network links in element *pointWithinLinearGNElement*.

Equipment description (CZ_DRU_MST_WEA, 12)

Element *measurementSiteRecord* contains a textual description of the equipment.

The equipment (weather station) has 8 capabilities defined. They are:

- precipitation,
- road surface conditions (3x),
- temperature,
- humidity,
- visibility and
- wind.

The road surface conditions is there 3 times (indexes 2-4). First one defines overall characteristics and is used to convey road status. Remaining two are road lane related measurements (one for each lane).

Element *measurementSiteLocation* of type *xsi:type="Point"* contains individual location description methods, in our case just one the description using Global Network links in element *pointWithinLinearGNElement*.

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_IntermediatePointOnLinearElement">
9     <xs:sequence>
10      <xs:element name="referent" type="D2LogicalModel:Referent" minOccurs="1"
  ↳ maxOccurs="1" />
11    </xs:sequence>
12    <xs:attribute name="index" type="xs:int" use="required" />
13  </xs:complexType>
14  <xs:complexType name="_LinearExtensionType">
15    <xs:sequence>
16      <xs:element name="openlrExtendedLinear" type=
  ↳ "D2LogicalModel:OpenlrExtendedLinear" minOccurs="0" />
17      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
18    </xs:sequence>
19  </xs:complexType>
20  <xs:complexType name="_MeasurementSiteRecordIndexMeasurementSpecificCharacteristics
  ↳ ">
21    <xs:sequence>
22      <xs:element name="measurementSpecificCharacteristics" type=
  ↳ "D2LogicalModel:MeasurementSpecificCharacteristics" minOccurs="1" maxOccurs="1" />
23    </xs:sequence>
24    <xs:attribute name="index" type="xs:int" use="required" />
25  </xs:complexType>
26  <xs:complexType name="_PointExtensionType">
27    <xs:sequence>
28      <xs:element name="openlrExtendedPoint" type="D2LogicalModel:OpenlrExtendedPoint
  ↳ " minOccurs="0" />
29      <xs:element name="pointExtended" type="D2LogicalModel:PointExtended" minOccurs=
  ↳ "0" />

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30     <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
    ↪ "unbounded" />
31   </xs:sequence>
32 </xs:complexType>
33 <xs:complexType name="_PredefinedLocationVersionedReference">
34   <xs:complexContent>
35     <xs:extension base="D2LogicalModel:VersionedReference">
36       <xs:attribute name="targetClass" use="required" fixed="PredefinedLocation" />
37     </xs:extension>
38   </xs:complexContent>
39 </xs:complexType>
40 <xs:complexType name="_VehicleCharacteristicsExtensionType">
41   <xs:sequence>
42     <xs:element name="vehicleCharacteristicsExtended" type=
    ↪ "D2LogicalModel:VehicleCharacteristicsExtended" minOccurs="0" />
43     <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
    ↪ "unbounded" />
44   </xs:sequence>
45 </xs:complexType>
46 <xs:complexType name="AffectedCarriagewayAndLanes">
47   <xs:sequence>
48     <xs:element name="carriageway" type="D2LogicalModel:CarriagewayEnum" minOccurs=
    ↪ "1" maxOccurs="1" />
49     <xs:element name="lane" type="D2LogicalModel:LaneEnum" minOccurs="0" maxOccurs=
    ↪ "unbounded" />
50     <xs:element name="footpath" type="D2LogicalModel:Boolean" minOccurs="0"
    ↪ maxOccurs="1" />
51     <xs:element name="lengthAffected" type="D2LogicalModel:MetresAsFloat" minOccurs=
    ↪ "0" maxOccurs="1" />
52     <xs:element name="affectedCarriagewayAndLanesExtension" type="D2LogicalModel:_
    ↪ ExtensionType" minOccurs="0" />
53   </xs:sequence>
54 </xs:complexType>
55 <xs:complexType name="AlertCDirection">
56   <xs:sequence>
57     <xs:element name="alertCDirectionCoded" type="D2LogicalModel:AlertCDirectionEnum
    ↪ " minOccurs="1" maxOccurs="1" />
58     <xs:element name="alertCDirectionNamed" type="D2LogicalModel:MultilingualString
    ↪ " minOccurs="0" maxOccurs="1" />
59     <xs:element name="alertCDirectionSense" type="D2LogicalModel:Boolean" minOccurs=
    ↪ "0" maxOccurs="1" />
60     <xs:element name="alertCDirectionExtension" type="D2LogicalModel:_ExtensionType
    ↪ " minOccurs="0" />
61   </xs:sequence>
62 </xs:complexType>
63 <xs:simpleType name="AlertCDirectionEnum">
64   <xs:restriction base="xs:string">
65     <xs:enumeration value="both" />
66     <xs:enumeration value="negative" />
67     <xs:enumeration value="positive" />
68     <xs:enumeration value="unknown" />
69   </xs:restriction>
70 </xs:simpleType>
71 <xs:complexType name="AlertCLinear" abstract="true">
72   <xs:sequence>
73     <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"

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74  <minOccurs="1" maxOccurs="1" />
    <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
75  <minOccurs="1" maxOccurs="1" />
    <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
76  <minOccurs="1" maxOccurs="1" />
    <xs:element name="alertCLinearExtension" type="D2LogicalModel:_ExtensionType"
77  <minOccurs="0" />
    </xs:sequence>
78  </xs:complexType>
79  <xs:complexType name="AlertCLinearByCode">
80  <xs:complexContent>
81  <xs:extension base="D2LogicalModel:AlertCLinear">
82  <xs:sequence>
83  <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
84  <xs:element name="locationCodeForLinearLocation" type=
85  <xs:element name="alertCLinearByCodeExtension" type="D2LogicalModel:_
86  <ExtensionType" minOccurs="0" />
    </xs:sequence>
87  </xs:extension>
88  </xs:complexContent>
89  </xs:complexType>
90  <xs:complexType name="AlertCLocation">
91  <xs:sequence>
92  <xs:element name="alertCLocationName" type="D2LogicalModel:MultilingualString"
93  <minOccurs="0" maxOccurs="1" />
    <xs:element name="specificLocation" type="D2LogicalModel:AlertCLocationCode"
94  <minOccurs="1" maxOccurs="1" />
    <xs:element name="alertCLocationExtension" type="D2LogicalModel:_ExtensionType"
95  <minOccurs="0" />
    </xs:sequence>
96  </xs:complexType>
97  <xs:simpleType name="AlertCLocationCode">
98  <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
99  </xs:simpleType>
100 <xs:complexType name="AlertCMethod2Linear">
101 <xs:complexContent>
102 <xs:extension base="D2LogicalModel:AlertCLinear">
103 <xs:sequence>
104 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
105 <xs:element name="alertCMethod2PrimaryPointLocation" type=
106 <xs:element name="alertCMethod2SecondaryPointLocation" type=
107 <xs:element name="alertCMethod2LinearExtension" type="D2LogicalModel:_
108 <ExtensionType" minOccurs="0" />
    </xs:sequence>
109 </xs:extension>
110 </xs:complexContent>
111 </xs:complexType>
112 <xs:complexType name="AlertCMethod2Point">
113 <xs:complexContent>
114 <xs:extension base="D2LogicalModel:AlertCPoint">
115 <xs:sequence>
116 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
117 <xs:element name="alertCMethod2PrimaryPointLocation" type=

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118 <xs:element name="alertCMethod2PrimaryPointLocation" />
119   <xs:sequence>
120     <xs:element name="alertCMethod2PointExtension" type="D2LogicalModel:_
121       ExtensionType" minOccurs="0" />
122   </xs:sequence>
123 </xs:extension>
124 </xs:complexType>
125 <xs:complexType name="AlertCMethod2PrimaryPointLocation">
126   <xs:sequence>
127     <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
128     <xs:element name="alertCMethod2PrimaryPointLocationExtension" type=
129       "D2LogicalModel:_ExtensionType" minOccurs="0" />
130   </xs:sequence>
131 </xs:complexType>
132 <xs:complexType name="AlertCMethod2SecondaryPointLocation">
133   <xs:sequence>
134     <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
135     <xs:element name="alertCMethod2SecondaryPointLocationExtension" type=
136       "D2LogicalModel:_ExtensionType" minOccurs="0" />
137   </xs:sequence>
138 </xs:complexType>
139 <xs:complexType name="AlertCMethod4Linear">
140   <xs:complexContent>
141     <xs:extension base="D2LogicalModel:AlertCLinear">
142       <xs:sequence>
143         <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
144         <xs:element name="alertCMethod4PrimaryPointLocation" type=
145           "D2LogicalModel:AlertCMethod4PrimaryPointLocation" />
146         <xs:element name="alertCMethod4SecondaryPointLocation" type=
147           "D2LogicalModel:AlertCMethod4SecondaryPointLocation" />
148         <xs:element name="alertCMethod4LinearExtension" type="D2LogicalModel:_
149           ExtensionType" minOccurs="0" />
150       </xs:sequence>
151     </xs:extension>
152   </xs:complexContent>
153 </xs:complexType>
154 <xs:complexType name="AlertCMethod4Point">
155   <xs:complexContent>
156     <xs:extension base="D2LogicalModel:AlertCPoint">
157       <xs:sequence>
158         <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
159         <xs:element name="alertCMethod4PrimaryPointLocation" type=
160           "D2LogicalModel:AlertCMethod4PrimaryPointLocation" />
161         <xs:element name="alertCMethod4PointExtension" type="D2LogicalModel:_
162           ExtensionType" minOccurs="0" />
163       </xs:sequence>
164     </xs:extension>
165   </xs:complexContent>
166 </xs:complexType>
167 <xs:complexType name="AlertCMethod4PrimaryPointLocation">
168   <xs:sequence>
169     <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
170     <xs:element name="offsetDistance" type="D2LogicalModel:OffsetDistance" />
171     <xs:element name="alertCMethod4PrimaryPointLocationExtension" type=
172       "D2LogicalModel:_ExtensionType" minOccurs="0" />
173   </xs:sequence>

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164 </xs:complexType>
165 <xs:complexType name="AlertCMethod4SecondaryPointLocation">
166   <xs:sequence>
167     <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
168     <xs:element name="offsetDistance" type="D2LogicalModel:OffsetDistance" />
169     <xs:element name="alertCMethod4SecondaryPointLocationExtension" type=
170     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
171   </xs:sequence>
172 </xs:complexType>
173 <xs:complexType name="AlertCPoint" abstract="true">
174   <xs:sequence>
175     <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String" ↪
176     ↪ minOccurs="1" maxOccurs="1" />
177     <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String" ↪
178     ↪ minOccurs="1" maxOccurs="1" />
179     <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String" ↪
180     ↪ minOccurs="1" maxOccurs="1" />
181     <xs:element name="alertCPointExtension" type="D2LogicalModel:_ExtensionType" ↪
182     ↪ minOccurs="0" />
183   </xs:sequence>
184 </xs:complexType>
185 <xs:simpleType name="AngleInDegrees">
186   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
187 </xs:simpleType>
188 <xs:complexType name="Area">
189   <xs:complexContent>
190     <xs:extension base="D2LogicalModel:Location">
191       <xs:sequence>
192         <xs:element name="areaExtension" type="D2LogicalModel:_ExtensionType" ↪
193         ↪ minOccurs="0" />
194       </xs:sequence>
195     </xs:extension>
196   </xs:complexContent>
197 </xs:complexType>
198 <xs:complexType name="AreaDestination">
199   <xs:complexContent>
200     <xs:extension base="D2LogicalModel:Destination">
201       <xs:sequence>
202         <xs:element name="area" type="D2LogicalModel:Area" />
203         <xs:element name="areaDestinationExtension" type="D2LogicalModel:_
204         ↪ ExtensionType" minOccurs="0" />
205       </xs:sequence>
206     </xs:extension>
207   </xs:complexContent>
208 </xs:complexType>
209 <xs:simpleType name="AreaOfInterestEnum">
210   <xs:restriction base="xs:string">
211     <xs:enumeration value="continentWide" />
212     <xs:enumeration value="national" />
213     <xs:enumeration value="neighbouringCountries" />
214     <xs:enumeration value="notSpecified" />
215     <xs:enumeration value="regional" />
216   </xs:restriction>
217 </xs:simpleType>
218 <xs:simpleType name="Boolean">
219   <xs:restriction base="xs:boolean" />

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213 </xs:simpleType>
214 <xs:simpleType name="CarriagewayEnum">
215   <xs:restriction base="xs:string">
216     <xs:enumeration value="connectingCarriageway" />
217     <xs:enumeration value="entrySlipRoad" />
218     <xs:enumeration value="exitSlipRoad" />
219     <xs:enumeration value="flyover" />
220     <xs:enumeration value="leftHandFeederRoad" />
221     <xs:enumeration value="leftHandParallelCarriageway" />
222     <xs:enumeration value="mainCarriageway" />
223     <xs:enumeration value="oppositeCarriageway" />
224     <xs:enumeration value="parallelCarriageway" />
225     <xs:enumeration value="rightHandFeederRoad" />
226     <xs:enumeration value="rightHandParallelCarriageway" />
227     <xs:enumeration value="roundabout" />
228     <xs:enumeration value="serviceRoad" />
229     <xs:enumeration value="slipRoads" />
230     <xs:enumeration value="underpass" />
231   </xs:restriction>
232 </xs:simpleType>
233 <xs:simpleType name="ComparisonOperatorEnum">
234   <xs:restriction base="xs:string">
235     <xs:enumeration value="equalTo" />
236     <xs:enumeration value="greaterThan" />
237     <xs:enumeration value="greaterThanOrEqualTo" />
238     <xs:enumeration value="lessThan" />
239     <xs:enumeration value="lessThanOrEqualTo" />
240   </xs:restriction>
241 </xs:simpleType>
242 <xs:simpleType name="ComputationMethodEnum">
243   <xs:restriction base="xs:string">
244     <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
245     <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
246     <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
247     <xs:enumeration value="medianOfSamplesInATimePeriod" />
248     <xs:enumeration value="movingAverageOfSamples" />
249   </xs:restriction>
250 </xs:simpleType>
251 <xs:simpleType name="ConfidentialityValueEnum">
252   <xs:restriction base="xs:string">
253     <xs:enumeration value="internalUse" />
254     <xs:enumeration value="noRestriction" />
255     <xs:enumeration value="restrictedToAuthorities" />
256     <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
257     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
258     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
259   </xs:restriction>
260 </xs:simpleType>
261 <xs:simpleType name="CountryEnum">
262   <xs:restriction base="xs:string">
263     <xs:enumeration value="at" />
264     <xs:enumeration value="be" />
265     <xs:enumeration value="bg" />
266     <xs:enumeration value="ch" />
267     <xs:enumeration value="cs" />

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268     <xs:enumeration value="cy" />
269     <xs:enumeration value="cz" />
270     <xs:enumeration value="de" />
271     <xs:enumeration value="dk" />
272     <xs:enumeration value="ee" />
273     <xs:enumeration value="es" />
274     <xs:enumeration value="fi" />
275     <xs:enumeration value="fo" />
276     <xs:enumeration value="fr" />
277     <xs:enumeration value="gb" />
278     <xs:enumeration value="gg" />
279     <xs:enumeration value="gi" />
280     <xs:enumeration value="gr" />
281     <xs:enumeration value="hr" />
282     <xs:enumeration value="hu" />
283     <xs:enumeration value="ie" />
284     <xs:enumeration value="im" />
285     <xs:enumeration value="is" />
286     <xs:enumeration value="it" />
287     <xs:enumeration value="je" />
288     <xs:enumeration value="li" />
289     <xs:enumeration value="lt" />
290     <xs:enumeration value="lu" />
291     <xs:enumeration value="lv" />
292     <xs:enumeration value="ma" />
293     <xs:enumeration value="mc" />
294     <xs:enumeration value="mk" />
295     <xs:enumeration value="mt" />
296     <xs:enumeration value="nl" />
297     <xs:enumeration value="no" />
298     <xs:enumeration value="pl" />
299     <xs:enumeration value="pt" />
300     <xs:enumeration value="ro" />
301     <xs:enumeration value="se" />
302     <xs:enumeration value="si" />
303     <xs:enumeration value="sk" />
304     <xs:enumeration value="sm" />
305     <xs:enumeration value="tr" />
306     <xs:enumeration value="va" />
307     <xs:enumeration value="other" />
308   </xs:restriction>
309 </xs:simpleType>
310 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
311   <xs:unique name="_d2LogicalModelMeasurementSiteRecordConstraint">
312     <xs:selector xpath="./D2LogicalModel:measurementSiteRecord" />
313     <xs:field xpath="@id" />
314     <xs:field xpath="@version" />
315   </xs:unique>
316   <xs:unique name="_d2LogicalModelMeasurementSiteTableConstraint">
317     <xs:selector xpath="./D2LogicalModel:measurementSiteTable" />
318     <xs:field xpath="@id" />
319     <xs:field xpath="@version" />
320   </xs:unique>
321 </xs:element>
322 <xs:complexType name="D2LogicalModel">
323   <xs:sequence>

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324     <xs:element name="exchange" type="D2LogicalModel:Exchange" />
325     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
↪ minOccurs="0" />
326     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
327   </xs:sequence>
328   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
329 </xs:complexType>
330 <xs:complexType name="DataValue" abstract="true">
331   <xs:sequence>
332     <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"
↪ maxOccurs="1" />
333     <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"
↪ minOccurs="0" maxOccurs="1" />
334     <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
335   </xs:sequence>
336   <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
337   <xs:attribute name="computationalMethod" type=
↪ "D2LogicalModel:ComputationMethodEnum" use="optional" />
338   <xs:attribute name="numberOfIncompleteInputs" type=
↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
339   <xs:attribute name="numberOfInputValuesUsed" type=
↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
340   <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
341   <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional"
↪ />
342   <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
↪ " use="optional" />
343 </xs:complexType>
344 <xs:simpleType name="DateTime">
345   <xs:restriction base="xs:dateTime" />
346 </xs:simpleType>
347 <xs:complexType name="Destination" abstract="true">
348   <xs:sequence>
349     <xs:element name="destinationExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
350   </xs:sequence>
351 </xs:complexType>
352 <xs:simpleType name="DirectionEnum">
353   <xs:restriction base="xs:string">
354     <xs:enumeration value="allDirections" />
355     <xs:enumeration value="bothWays" />
356     <xs:enumeration value="clockwise" />
357     <xs:enumeration value="anticlockwise" />
358     <xs:enumeration value="innerRing" />
359     <xs:enumeration value="outerRing" />
360     <xs:enumeration value="northBound" />
361     <xs:enumeration value="northEastBound" />
362     <xs:enumeration value="eastBound" />
363     <xs:enumeration value="southEastBound" />
364     <xs:enumeration value="southBound" />
365     <xs:enumeration value="southWestBound" />
366     <xs:enumeration value="westBound" />
367     <xs:enumeration value="northWestBound" />
368     <xs:enumeration value="inboundTowardsTown" />

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369     <xs:enumeration value="outboundFromTown" />
370     <xs:enumeration value="unknown" />
371     <xs:enumeration value="opposite" />
372     <xs:enumeration value="other" />
373   </xs:restriction>
374 </xs:simpleType>
375 <xs:complexType name="DistanceAlongLinearElement" abstract="true">
376   <xs:sequence>
377     <xs:element name="distanceAlongLinearElementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
378   </xs:sequence>
379 </xs:complexType>
380 <xs:complexType name="DistanceFromLinearElementReferent">
381   <xs:complexContent>
382     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
383       <xs:sequence>
384         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
↪ minOccurs="1" maxOccurs="1" />
385         <xs:element name="fromReferent" type="D2LogicalModel:Referent" />
386         <xs:element name="towardsReferent" type="D2LogicalModel:Referent" minOccurs=
↪ "0" />
387         <xs:element name="distanceFromLinearElementReferentExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
388       </xs:sequence>
389     </xs:extension>
390   </xs:complexContent>
391 </xs:complexType>
392 <xs:complexType name="DistanceFromLinearElementStart">
393   <xs:complexContent>
394     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
395       <xs:sequence>
396         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
↪ minOccurs="1" maxOccurs="1" />
397         <xs:element name="distanceFromLinearElementStartExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
398       </xs:sequence>
399     </xs:extension>
400   </xs:complexContent>
401 </xs:complexType>
402 <xs:complexType name="ElaboratedDataFault">
403   <xs:complexContent>
404     <xs:extension base="D2LogicalModel:Fault">
405       <xs:sequence>
406         <xs:element name="elaboratedDataFault" type=
↪ "D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
407         <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
408       </xs:sequence>
409     </xs:extension>
410   </xs:complexContent>
411 </xs:complexType>
412 <xs:simpleType name="ElaboratedDataFaultEnum">
413   <xs:restriction base="xs:string">
414     <xs:enumeration value="intermittentDataValues" />
415     <xs:enumeration value="noDataValuesAvailable" />
416     <xs:enumeration value="spuriousUnreliableDataValues" />

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417     <xs:enumeration value="unspecifiedOrUnknownFault" />
418     <xs:enumeration value="other" />
419   </xs:restriction>
420 </xs:simpleType>
421 <xs:complexType name="Exchange">
422   <xs:sequence>
423     <xs:element name="supplierIdentification" type=
↪ "D2LogicalModel:InternationalIdentifier" />
424     <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType" ↪
↪ minOccurs="0" />
425   </xs:sequence>
426 </xs:complexType>
427 <xs:complexType name="Fault">
428   <xs:sequence>
429     <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0" ↪
↪ maxOccurs="1" />
430     <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0" ↪
↪ maxOccurs="1" />
431     <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0" ↪
↪ maxOccurs="1" />
432     <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
↪ "1" maxOccurs="1" />
433     <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum" ↪
↪ minOccurs="0" maxOccurs="1" />
434     <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType" ↪
↪ minOccurs="0" />
435   </xs:sequence>
436 </xs:complexType>
437 <xs:simpleType name="FaultSeverityEnum">
438   <xs:restriction base="xs:string">
439     <xs:enumeration value="low" />
440     <xs:enumeration value="medium" />
441     <xs:enumeration value="high" />
442     <xs:enumeration value="unknown" />
443   </xs:restriction>
444 </xs:simpleType>
445 <xs:simpleType name="Float">
446   <xs:restriction base="xs:float" />
447 </xs:simpleType>
448 <xs:simpleType name="FuelType2Enum">
449   <xs:restriction base="xs:string">
450     <xs:enumeration value="all" />
451     <xs:enumeration value="petrol95Octane" />
452     <xs:enumeration value="petrol98Octane" />
453     <xs:enumeration value="petrolLeaded" />
454     <xs:enumeration value="petrolUnleaded" />
455     <xs:enumeration value="unknown" />
456     <xs:enumeration value="other" />
457   </xs:restriction>
458 </xs:simpleType>
459 <xs:simpleType name="FuelTypeEnum">
460   <xs:restriction base="xs:string">
461     <xs:enumeration value="battery" />
462     <xs:enumeration value="biodiesel" />
463     <xs:enumeration value="diesel" />
464     <xs:enumeration value="dieselBatteryHybrid" />

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465     <xs:enumeration value="ethanol" />
466     <xs:enumeration value="hydrogen" />
467     <xs:enumeration value="liquidGas" />
468     <xs:enumeration value="lpg" />
469     <xs:enumeration value="methane" />
470     <xs:enumeration value="petrol" />
471     <xs:enumeration value="petrolBatteryHybrid" />
472   </xs:restriction>
473 </xs:simpleType>
474 <xs:simpleType name="GeometryTypeEnum">
475   <xs:restriction base="xs:string" />
476 </xs:simpleType>
477 <xs:complexType name="GlobalNetworkLinear">
478   <xs:sequence>
479     <xs:element name="networkVersion" type="D2LogicalModel:MultilingualString"
480     ↪ minOccurs="1" maxOccurs="1" />
481     <xs:element name="linearGeometryType" type="D2LogicalModel:GeometryTypeEnum"
482     ↪ minOccurs="1" maxOccurs="1" />
483     <xs:element name="startPoint" type="D2LogicalModel:PointByMultiCoordinates"
484     ↪ minOccurs="0" />
485     <xs:element name="endPoint" type="D2LogicalModel:PointByMultiCoordinates"
486     ↪ minOccurs="0" />
487     <xs:element name="linearWithinLinearGNElement" type=
488     ↪ "D2LogicalModel:LinearWithinLinearGNElement" maxOccurs="unbounded" />
489   </xs:sequence>
490 </xs:complexType>
491 <xs:complexType name="GrossWeightCharacteristic">
492   <xs:sequence>
493     <xs:element name="comparisonOperator" type=
494     ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
495     <xs:element name="grossVehicleWeight" type="D2LogicalModel:Tonnes" minOccurs="1"
496     ↪ " maxOccurs="1" />
497     <xs:element name="grossWeightCharacteristicExtension" type="D2LogicalModel:_
498     ↪ ExtensionType" minOccurs="0" />
499   </xs:sequence>
500 </xs:complexType>
501 <xs:complexType name="GroupOfLocations" abstract="true">
502   <xs:sequence>
503     <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType"
504     ↪ " minOccurs="0" />
505   </xs:sequence>
506 </xs:complexType>
507 <xs:complexType name="HeaderInformation">
508   <xs:sequence>
509     <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum"
510     ↪ minOccurs="0" maxOccurs="1" />
511     <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum"
512     ↪ " minOccurs="1" maxOccurs="1" />
513     <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum"
514     ↪ " minOccurs="1" maxOccurs="1" />
515     <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
516     ↪ maxOccurs="1" />
517     <xs:element name="headerInformationExtension" type="D2LogicalModel:_
518     ↪ ExtensionType" minOccurs="0" />
519   </xs:sequence>
520 </xs:complexType>

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507 <xs:complexType name="HeaviestAxleWeightCharacteristic">
508   <xs:sequence>
509     <xs:element name="comparisonOperator" type=
510     ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
511     <xs:element name="heaviestAxleWeight" type="D2LogicalModel:Tonnes" minOccurs="1"
512     ↪ " maxOccurs="1" />
513     <xs:element name="heaviestAxleWeightCharacteristicExtension" type=
514     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
515   </xs:sequence>
516 </xs:complexType>
517 <xs:complexType name="HeightCharacteristic">
518   <xs:sequence>
519     <xs:element name="comparisonOperator" type=
520     ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
521     <xs:element name="vehicleHeight" type="D2LogicalModel:MetresAsFloat" minOccurs=
522     ↪ "1" maxOccurs="1" />
523     <xs:element name="heightCharacteristicExtension" type="D2LogicalModel:_
524     ↪ ExtensionType" minOccurs="0" />
525   </xs:sequence>
526 </xs:complexType>
527 <xs:simpleType name="HeightGradeEnum">
528   <xs:restriction base="xs:string">
529     <xs:enumeration value="aboveGrade" />
530     <xs:enumeration value="atGrade" />
531     <xs:enumeration value="belowGrade" />
532   </xs:restriction>
533 </xs:simpleType>
534 <xs:simpleType name="InformationStatusEnum">
535   <xs:restriction base="xs:string">
536     <xs:enumeration value="real" />
537     <xs:enumeration value="securityExercise" />
538     <xs:enumeration value="technicalExercise" />
539     <xs:enumeration value="test" />
540   </xs:restriction>
541 </xs:simpleType>
542 <xs:simpleType name="Integer">
543   <xs:restriction base="xs:integer" />
544 </xs:simpleType>
545 <xs:complexType name="InternationalIdentifier">
546   <xs:sequence>
547     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
548     ↪ maxOccurs="1" />
549     <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1"
550     ↪ " maxOccurs="1" />
551     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
552     ↪ ExtensionType" minOccurs="0" />
553   </xs:sequence>
554 </xs:complexType>
555 <xs:complexType name="Junction">
556   <xs:sequence>
557     <xs:element name="junctionClassification" type=
558     ↪ "D2LogicalModel:JunctionClassificationEnum" minOccurs="0" maxOccurs="1" />
559     <xs:element name="junctionName" type="D2LogicalModel:MultilingualString"
560     ↪ minOccurs="1" maxOccurs="1" />
561     <xs:element name="junctionNumber" type="D2LogicalModel:String" minOccurs="0"
562     ↪ maxOccurs="1" />

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551     <xs:element name="motorway" type="D2LogicalModel:Road" minOccurs="0" />
552     <xs:element name="destinationMotorway" type="D2LogicalModel:Road" minOccurs="0"
↪maxOccurs="unbounded" />
553     <xs:element name="junctionExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />
554   </xs:sequence>
555 </xs:complexType>
556 <xs:simpleType name="JunctionClassificationEnum">
557   <xs:restriction base="xs:string">
558     <xs:enumeration value="threeWayInterchange" />
559     <xs:enumeration value="interchange" />
560     <xs:enumeration value="motorwayConnection" />
561     <xs:enumeration value="junction" />
562     <xs:enumeration value="temporaryJunction" />
563     <xs:enumeration value="borderCrossing" />
564     <xs:enumeration value="junctionInOneDirection" />
565     <xs:enumeration value="operationalServiceJunction" />
566     <xs:enumeration value="other" />
567   </xs:restriction>
568 </xs:simpleType>
569 <xs:simpleType name="LaneEnum">
570   <xs:restriction base="xs:string">
571     <xs:enumeration value="allLanesCompleteCarriageway" />
572     <xs:enumeration value="busLane" />
573     <xs:enumeration value="busStop" />
574     <xs:enumeration value="carPoolLane" />
575     <xs:enumeration value="centralReservation" />
576     <xs:enumeration value="crawlerLane" />
577     <xs:enumeration value="emergencyLane" />
578     <xs:enumeration value="escapeLane" />
579     <xs:enumeration value="expressLane" />
580     <xs:enumeration value="hardShoulder" />
581     <xs:enumeration value="heavyVehicleLane" />
582     <xs:enumeration value="lane1" />
583     <xs:enumeration value="lane2" />
584     <xs:enumeration value="lane3" />
585     <xs:enumeration value="lane4" />
586     <xs:enumeration value="lane5" />
587     <xs:enumeration value="lane6" />
588     <xs:enumeration value="lane7" />
589     <xs:enumeration value="lane8" />
590     <xs:enumeration value="lane9" />
591     <xs:enumeration value="layBy" />
592     <xs:enumeration value="leftHandTurningLane" />
593     <xs:enumeration value="leftLane" />
594     <xs:enumeration value="localTrafficLane" />
595     <xs:enumeration value="middleLane" />
596     <xs:enumeration value="opposingLanes" />
597     <xs:enumeration value="overtakingLane" />
598     <xs:enumeration value="rightHandTurningLane" />
599     <xs:enumeration value="rightLane" />
600     <xs:enumeration value="rushHourLane" />
601     <xs:enumeration value="setDownArea" />
602     <xs:enumeration value="slowVehicleLane" />
603     <xs:enumeration value="throughTrafficLane" />
604     <xs:enumeration value="tidalFlowLane" />

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605     <xs:enumeration value="turningLane" />
606     <xs:enumeration value="verge" />
607   </xs:restriction>
608 </xs:simpleType>
609 <xs:simpleType name="Language">
610   <xs:restriction base="xs:language" />
611 </xs:simpleType>
612 <xs:complexType name="LengthCharacteristic">
613   <xs:sequence>
614     <xs:element name="comparisonOperator" type=
615 ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
616     <xs:element name="vehicleLength" type="D2LogicalModel:MetresAsFloat" minOccurs=
617 ↪ "1" maxOccurs="1" />
618     <xs:element name="lengthCharacteristicExtension" type="D2LogicalModel:_
619 ↪ ExtensionType" minOccurs="0" />
620   </xs:sequence>
621 </xs:complexType>
622 <xs:complexType name="Linear">
623   <xs:complexContent>
624     <xs:extension base="D2LogicalModel:NetworkLocation">
625       <xs:sequence>
626         <xs:element name="tpegLinearLocation" type=
627 ↪ "D2LogicalModel:TpegLinearLocation" minOccurs="0" />
628         <xs:element name="alertCLinear" type="D2LogicalModel:AlertCLinear"
629 ↪ minOccurs="0" />
630         <xs:element name="linearWithinLinearElement" type=
631 ↪ "D2LogicalModel:LinearWithinLinearElement" minOccurs="0" />
632         <xs:element name="globalNetworkLinear" type=
633 ↪ "D2LogicalModel:GlobalNetworkLinear" minOccurs="0" />
634         <xs:element name="linearExtension" type="D2LogicalModel:_LinearExtensionType
635 ↪ " minOccurs="0" />
636       </xs:sequence>
637     </xs:extension>
638   </xs:complexContent>
639 </xs:complexType>
640 <xs:complexType name="LinearElement">
641   <xs:sequence>
642     <xs:element name="roadName" type="D2LogicalModel:MultilingualString" minOccurs=
643 ↪ "0" maxOccurs="1" />
644     <xs:element name="roadNumber" type="D2LogicalModel:String" minOccurs="0"
645 ↪ maxOccurs="1" />
646     <xs:element name="linearElementReferenceModel" type="D2LogicalModel:String"
647 ↪ minOccurs="0" maxOccurs="1" />
648     <xs:element name="linearElementReferenceModelVersion" type=
649 ↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
650     <xs:element name="linearElementNature" type=
651 ↪ "D2LogicalModel:LinearElementNatureEnum" minOccurs="0" maxOccurs="1" />
652     <xs:element name="linearElementExtension" type="D2LogicalModel:_ExtensionType"
653 ↪ minOccurs="0" />
654   </xs:sequence>
655 </xs:complexType>
656 <xs:complexType name="LinearElementByCode">
657   <xs:complexContent>
658     <xs:extension base="D2LogicalModel:LinearElement">
659       <xs:sequence>
660         <xs:element name="linearElementIdentifier" type="D2LogicalModel:String"

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647   minOccurs="1" maxOccurs="1" />
        <xs:element name="linearElementByCodeExtension" type="D2LogicalModel:_
648   ExtensionType" minOccurs="0" />
        </xs:sequence>
649   </xs:extension>
650   </xs:complexContent>
651 </xs:complexType>
652 <xs:complexType name="LinearElementByPoints">
653   <xs:complexContent>
654     <xs:extension base="D2LogicalModel:LinearElement">
655       <xs:sequence>
656         <xs:element name="startPointOfLinearElement" type="D2LogicalModel:Referent"
657   </>
        <xs:element name="intermediatePointOnLinearElement" type="D2LogicalModel:_
658   IntermediatePointOnLinearElement" minOccurs="0" maxOccurs="unbounded" />
        <xs:element name="endPointOfLinearElement" type="D2LogicalModel:Referent" />
659         <xs:element name="linearElementByPointsExtension" type="D2LogicalModel:_
660   ExtensionType" minOccurs="0" />
        </xs:sequence>
661     </xs:extension>
662   </xs:complexContent>
663 </xs:complexType>
664 <xs:simpleType name="LinearElementNatureEnum">
665   <xs:restriction base="xs:string">
666     <xs:enumeration value="road" />
667     <xs:enumeration value="roadSection" />
668     <xs:enumeration value="slipRoad" />
669     <xs:enumeration value="other" />
670   </xs:restriction>
671 </xs:simpleType>
672 <xs:simpleType name="LinearReferencingDirectionEnum">
673   <xs:restriction base="xs:string">
674     <xs:enumeration value="both" />
675     <xs:enumeration value="opposite" />
676     <xs:enumeration value="aligned" />
677     <xs:enumeration value="unknown" />
678   </xs:restriction>
679 </xs:simpleType>
680 <xs:complexType name="LinearWithinLinearElement">
681   <xs:sequence>
682     <xs:element name="administrativeAreaOfLinearSection" type=
683   "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
        <xs:element name="directionBoundOnLinearSection" type=
684   "D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="1" />
        <xs:element name="directionRelativeOnLinearSection" type=
685   "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
        <xs:element name="heightGradeOfLinearSection" type=
686   "D2LogicalModel:HeightGradeEnum" minOccurs="0" maxOccurs="1" />
        <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
687     <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
688     <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
689     <xs:element name="linearWithinLinearElementExtension" type="D2LogicalModel:_
690   ExtensionType" minOccurs="0" />
        </xs:sequence>
691 </xs:complexType>
692 <xs:complexType name="LinearWithinLinearGNElement">

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693 <xs:sequence>
694   <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
695   ↪ "0" maxOccurs="1" />
696   <xs:element name="directionRelativeOnLinearSection" type=
697   ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
698   <xs:element name="orderOfSection" type="D2LogicalModel:NonNegativeInteger"
699   ↪ minOccurs="0" maxOccurs="1" />
700   <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
701   <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
702   <xs:element name="linearWithinLinearGNElementExtension" type="D2LogicalModel:_
703   ↪ ExtensionType" minOccurs="0" />
704 </xs:sequence>
705 </xs:complexType>
706 <xs:simpleType name="LoadType2Enum">
707   <xs:restriction base="xs:string">
708     <xs:enumeration value="refrigeratedGoods" />
709   </xs:restriction>
710 </xs:simpleType>
711 <xs:simpleType name="LoadTypeEnum">
712   <xs:restriction base="xs:string">
713     <xs:enumeration value="abnormalLoad" />
714     <xs:enumeration value="ammunition" />
715     <xs:enumeration value="chemicals" />
716     <xs:enumeration value="combustibleMaterials" />
717     <xs:enumeration value="corrosiveMaterials" />
718     <xs:enumeration value="debris" />
719     <xs:enumeration value="empty" />
720     <xs:enumeration value="explosiveMaterials" />
721     <xs:enumeration value="extraHighLoad" />
722     <xs:enumeration value="extraLongLoad" />
723     <xs:enumeration value="extraWideLoad" />
724     <xs:enumeration value="fuel" />
725     <xs:enumeration value="glass" />
726     <xs:enumeration value="goods" />
727     <xs:enumeration value="hazardousMaterials" />
728     <xs:enumeration value="liquid" />
729     <xs:enumeration value="livestock" />
730     <xs:enumeration value="materials" />
731     <xs:enumeration value="materialsDangerousForPeople" />
732     <xs:enumeration value="materialsDangerousForTheEnvironment" />
733     <xs:enumeration value="materialsDangerousForWater" />
734     <xs:enumeration value="oil" />
735     <xs:enumeration value="ordinary" />
736     <xs:enumeration value="perishableProducts" />
737     <xs:enumeration value="petrol" />
738     <xs:enumeration value="pharmaceuticalMaterials" />
739     <xs:enumeration value="radioactiveMaterials" />
740     <xs:enumeration value="refuse" />
741     <xs:enumeration value="toxicMaterials" />
742     <xs:enumeration value="vehicles" />
743     <xs:enumeration value="other" />
744   </xs:restriction>
</xs:simpleType>
<xs:complexType name="Location" abstract="true">
  <xs:complexContent>
    <xs:extension base="D2LogicalModel:GroupOfLocations">

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745     <xs:sequence>
746       <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
747     </xs:sequence>
748   </xs:extension>
749 </xs:complexContent>
750 </xs:complexType>
751 <xs:complexType name="LocationByReference">
752   <xs:complexContent>
753     <xs:extension base="D2LogicalModel:Location">
754       <xs:sequence>
755         <xs:element name="predefinedLocationReference" type="D2LogicalModel:_
↪ PredefinedLocationVersionedReference" minOccurs="1" maxOccurs="1" />
756         <xs:element name="locationByReferenceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
757       </xs:sequence>
758     </xs:extension>
759   </xs:complexContent>
760 </xs:complexType>
761 <xs:simpleType name="LocationDescriptorEnum">
762   <xs:restriction base="xs:string">
763     <xs:enumeration value="aroundABendInRoad" />
764     <xs:enumeration value="atMotorwayInterchange" />
765     <xs:enumeration value="atRestArea" />
766     <xs:enumeration value="atServiceArea" />
767     <xs:enumeration value="atTollPlaza" />
768     <xs:enumeration value="atTunnelEntryOrExit" />
769     <xs:enumeration value="inbound" />
770     <xs:enumeration value="inGallery" />
771     <xs:enumeration value="inTheCentre" />
772     <xs:enumeration value="inTheOppositeDirection" />
773     <xs:enumeration value="inTunnel" />
774     <xs:enumeration value="onBorder" />
775     <xs:enumeration value="onBridge" />
776     <xs:enumeration value="onConnector" />
777     <xs:enumeration value="onElevatedSection" />
778     <xs:enumeration value="onFlyover" />
779     <xs:enumeration value="onIceRoad" />
780     <xs:enumeration value="onLevelCrossing" />
781     <xs:enumeration value="onLinkRoad" />
782     <xs:enumeration value="onPass" />
783     <xs:enumeration value="onRoundabout" />
784     <xs:enumeration value="onTheLeft" />
785     <xs:enumeration value="onTheRight" />
786     <xs:enumeration value="onTheRoadway" />
787     <xs:enumeration value="onUndergroundSection" />
788     <xs:enumeration value="onUnderpass" />
789     <xs:enumeration value="outbound" />
790     <xs:enumeration value="overCrestOfHill" />
791     <xs:enumeration value="withinJunction" />
792   </xs:restriction>
793 </xs:simpleType>
794 <xs:simpleType name="MeasuredOrDerivedDataTypeEnum">
795   <xs:restriction base="xs:string">
796     <xs:enumeration value="humidityInformation" />
797     <xs:enumeration value="individualVehicleMeasurements" />

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798     <xs:enumeration value="pollutionInformation" />
799     <xs:enumeration value="precipitationInformation" />
800     <xs:enumeration value="pressureInformation" />
801     <xs:enumeration value="radiationInformation" />
802     <xs:enumeration value="roadSurfaceConditionInformation" />
803     <xs:enumeration value="temperatureInformation" />
804     <xs:enumeration value="trafficConcentration" />
805     <xs:enumeration value="trafficFlow" />
806     <xs:enumeration value="trafficHeadway" />
807     <xs:enumeration value="trafficSpeed" />
808     <xs:enumeration value="trafficStatusInformation" />
809     <xs:enumeration value="travelTimeInformation" />
810     <xs:enumeration value="visibilityInformation" />
811     <xs:enumeration value="windInformation" />
812   </xs:restriction>
813 </xs:simpleType>
814 <xs:complexType name="MeasurementSiteRecord">
815   <xs:sequence>
816     <xs:element name="measurementSiteRecordVersionTime" type=
817 ↪ "D2LogicalModel:DateTime" minOccurs="0" maxOccurs="1" />
818     <xs:element name="computationMethod" type="D2LogicalModel:ComputationMethodEnum"
819 ↪ minOccurs="0" maxOccurs="1" />
820     <xs:element name="measurementEquipmentReference" type="D2LogicalModel:String"
821 ↪ minOccurs="0" maxOccurs="1" />
822     <xs:element name="measurementEquipmentTypeUsed" type=
823 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
824     <xs:element name="measurementSiteName" type="D2LogicalModel:MultilingualString"
825 ↪ minOccurs="0" maxOccurs="1" />
826     <xs:element name="measurementSiteNumberOfLanes" type=
827 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
828     <xs:element name="measurementSiteIdentification" type="D2LogicalModel:String"
829 ↪ minOccurs="0" maxOccurs="1" />
830     <xs:element name="measurementSide" type="D2LogicalModel:DirectionEnum"
831 ↪ minOccurs="0" maxOccurs="1" />
832     <xs:element name="measurementSpecificCharacteristics" type="D2LogicalModel:_
833 ↪ MeasurementSiteRecordIndexMeasurementSpecificCharacteristics" minOccurs="0"
834 ↪ maxOccurs="unbounded" />
835     <xs:element name="measurementSiteLocation" type="D2LogicalModel:GroupOfLocations"
836 ↪ />
837     <xs:element name="measurementSiteRecordExtension" type="D2LogicalModel:_
838 ↪ ExtensionType" minOccurs="0" />
839   </xs:sequence>
840   <xs:attribute name="id" type="xs:string" use="required" />
841   <xs:attribute name="version" type="xs:string" use="required" />
842 </xs:complexType>
843 <xs:complexType name="MeasurementSiteTable">
844   <xs:sequence>
845     <xs:element name="measurementSiteTableIdentification" type=
846 ↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
847     <xs:element name="measurementSiteRecord" type=
848 ↪ "D2LogicalModel:MeasurementSiteRecord" maxOccurs="unbounded" />
849     <xs:element name="measurementSiteTableExtension" type="D2LogicalModel:_
850 ↪ ExtensionType" minOccurs="0" />
851   </xs:sequence>
852   <xs:attribute name="id" type="xs:string" use="required" />
853   <xs:attribute name="version" type="xs:string" use="required" />

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839 </xs:complexType>
840 <xs:complexType name="MeasurementSiteTablePublication">
841   <xs:complexContent>
842     <xs:extension base="D2LogicalModel:PayloadPublication">
843       <xs:sequence>
844         <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation
845 ↪ " />
846         <xs:element name="measurementSiteTable" type=
847 ↪ "D2LogicalModel:MeasurementSiteTable" maxOccurs="unbounded" />
848         <xs:element name="measurementSiteTablePublicationExtension" type=
849 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
850       </xs:sequence>
851     </xs:extension>
852   </xs:complexContent>
853 </xs:complexType>
854 <xs:complexType name="MeasurementSpecificCharacteristics">
855   <xs:sequence>
856     <xs:element name="accuracy" type="D2LogicalModel:Percentage" minOccurs="0"
857 ↪ maxOccurs="1" />
858     <xs:element name="period" type="D2LogicalModel:Seconds" minOccurs="0" maxOccurs=
859 ↪ "1" />
860     <xs:element name="smoothingFactor" type="D2LogicalModel:Float" minOccurs="0"
861 ↪ maxOccurs="1" />
862     <xs:element name="specificLane" type="D2LogicalModel:LaneEnum" minOccurs="0"
863 ↪ maxOccurs="1" />
864     <xs:element name="specificMeasurementValueType" type=
865 ↪ "D2LogicalModel:MeasuredOrDerivedDataTypeEnum" minOccurs="1" maxOccurs="1" />
866     <xs:element name="specificVehicleCharacteristics" type=
867 ↪ "D2LogicalModel:VehicleCharacteristics" minOccurs="0" />
868     <xs:element name="measurementSpecificCharacteristicsExtension" type=
869 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
870   </xs:sequence>
871 </xs:complexType>
872 <xs:simpleType name="MetresAsFloat">
873   <xs:restriction base="D2LogicalModel:Float" />
874 </xs:simpleType>
875 <xs:simpleType name="MetresAsNonNegativeInteger">
876   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
877 </xs:simpleType>
878 <xs:complexType name="MultilingualString">
879   <xs:sequence>
880     <xs:element name="values">
881       <xs:complexType>
882         <xs:sequence>
883           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
884 ↪ maxOccurs="unbounded" />
885         </xs:sequence>
886       </xs:complexType>
887     </xs:element>
888   </xs:sequence>
889 </xs:complexType>
890 <xs:complexType name="MultilingualStringValue">
891   <xs:simpleContent>
892     <xs:extension base="D2LogicalModel:MultilingualStringValue" type="
893 ↪ "D2LogicalModel:MultilingualStringValue" />
894     <xs:attribute name="lang" type="xs:language" />
895   </xs:extension>

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884     </xs:simpleContent>
885 </xs:complexType>
886 <xs:simpleType name="MultilingualStringValue">
887     <xs:restriction base="xs:string">
888         <xs:maxLength value="1024" />
889     </xs:restriction>
890 </xs:simpleType>
891 <xs:complexType name="NetworkLocation" abstract="true">
892     <xs:complexContent>
893         <xs:extension base="D2LogicalModel:Location">
894             <xs:sequence>
895                 <xs:element name="supplementaryPositionalDescription" type=
896 ↪ "D2LogicalModel:SupplementaryPositionalDescription" minOccurs="0" />
897                 <xs:element name="destination" type="D2LogicalModel:Destination" minOccurs=
898 ↪ "0" />
899                 <xs:element name="networkLocationExtension" type="D2LogicalModel:_
900 ↪ ExtensionType" minOccurs="0" />
901             </xs:sequence>
902         </xs:extension>
903     </xs:complexContent>
904 </xs:complexType>
905 <xs:simpleType name="NonNegativeInteger">
906     <xs:restriction base="xs:nonNegativeInteger" />
907 </xs:simpleType>
908 <xs:complexType name="NumberOfAxlesCharacteristic">
909     <xs:sequence>
910         <xs:element name="comparisonOperator" type=
911 ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
912         <xs:element name="numberOfAxles" type="D2LogicalModel:NonNegativeInteger"
913 ↪ minOccurs="1" maxOccurs="1" />
914         <xs:element name="numberOfAxlesCharacteristicExtension" type="D2LogicalModel:_
915 ↪ ExtensionType" minOccurs="0" />
916     </xs:sequence>
917 </xs:complexType>
918 <xs:complexType name="OccupancyChangeValue">
919     <xs:complexContent>
920         <xs:extension base="D2LogicalModel:DataValue">
921             <xs:sequence>
922                 <xs:element name="occupancyChange" type="D2LogicalModel:Integer" minOccurs=
923 ↪ "1" maxOccurs="1" />
924                 <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
925 ↪ ExtensionType" minOccurs="0" />
926             </xs:sequence>
927         </xs:extension>
928     </xs:complexContent>
929 </xs:complexType>
930 <xs:complexType name="OffsetDistance">
931     <xs:sequence>
932         <xs:element name="offsetDistance" type=
933 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="1" maxOccurs="1" />
934         <xs:element name="offsetDistanceExtension" type="D2LogicalModel:_ExtensionType"
935 ↪ minOccurs="0" />
936     </xs:sequence>
937 </xs:complexType>
938 <xs:complexType name="OpenlrBaseLocationReferencePoint" abstract="true">
939     <xs:sequence>

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930     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
931     <xs:element name="openlrLineAttributes" type=
932     ↪ "D2LogicalModel:OpenlrLineAttributes" />
933     <xs:element name="openlrBaseLocationReferencePointExtension" type=
934     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
935     </xs:sequence>
936   </xs:complexType>
937   <xs:complexType name="OpenlrBasePointLocation" abstract="true">
938     <xs:sequence>
939       <xs:element name="openlrSideOfRoad" type="D2LogicalModel:OpenlrSideOfRoadEnum"
940       ↪ minOccurs="1" maxOccurs="1" />
941       <xs:element name="openlrOrientation" type="D2LogicalModel:OpenlrOrientationEnum"
942       ↪ minOccurs="1" maxOccurs="1" />
943       <xs:element name="openlrPositiveOffset" type=
944       ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
945       <xs:element name="openlrLocationReferencePoint" type=
946       ↪ "D2LogicalModel:OpenlrLocationReferencePoint" />
947       <xs:element name="openlrLastLocationReferencePoint" type=
948       ↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
949       <xs:element name="openlrBasePointLocationExtension" type="D2LogicalModel:_
950       ↪ ExtensionType" minOccurs="0" />
951     </xs:sequence>
952   </xs:complexType>
953   <xs:complexType name="OpenlrExtendedLinear">
954     <xs:sequence>
955       <xs:element name="firstDirection" type=
956       ↪ "D2LogicalModel:OpenlrLineLocationReference" />
957       <xs:element name="oppositeDirection" type=
958       ↪ "D2LogicalModel:OpenlrLineLocationReference" minOccurs="0" />
959     </xs:sequence>
960   </xs:complexType>
961   <xs:complexType name="OpenlrExtendedPoint">
962     <xs:sequence>
963       <xs:element name="openlrPointLocationReference" type=
964       ↪ "D2LogicalModel:OpenlrPointLocationReference" />
965     </xs:sequence>
966   </xs:complexType>
967   <xs:simpleType name="OpenlrFormOfWayEnum">
968     <xs:restriction base="xs:string">
969       <xs:enumeration value="undefined" />
970       <xs:enumeration value="motorway" />
971       <xs:enumeration value="multipleCarriageway" />
972       <xs:enumeration value="singleCarriageway" />
973       <xs:enumeration value="roundabout" />
974       <xs:enumeration value="slipRoad" />
975       <xs:enumeration value="trafficSquare" />
976       <xs:enumeration value="other" />
977     </xs:restriction>
978   </xs:simpleType>
979   <xs:simpleType name="OpenlrFunctionalRoadClassEnum">
980     <xs:restriction base="xs:string">
981       <xs:enumeration value="FRC0" />
982       <xs:enumeration value="FRC1" />
983       <xs:enumeration value="FRC2" />
984       <xs:enumeration value="FRC3" />
985       <xs:enumeration value="FRC4" />

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975     <xs:enumeration value="FRC5" />
976     <xs:enumeration value="FRC6" />
977     <xs:enumeration value="FRC7" />
978   </xs:restriction>
979 </xs:simpleType>
980 <xs:complexType name="OpenlrGeoCoordinate">
981   <xs:sequence>
982     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
983     <xs:element name="openlrGeoCoordinateExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
984   </xs:sequence>
985 </xs:complexType>
986 <xs:complexType name="OpenlrLastLocationReferencePoint">
987   <xs:complexContent>
988     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
989       <xs:sequence>
990         <xs:element name="openlrLastLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
991       </xs:sequence>
992     </xs:extension>
993   </xs:complexContent>
994 </xs:complexType>
995 <xs:complexType name="OpenlrLineAttributes">
996   <xs:sequence>
997     <xs:element name="openlrFunctionalRoadClass" type=
↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
998     <xs:element name="openlrFormOfWay" type="D2LogicalModel:OpenlrFormOfWayEnum"
↪ minOccurs="1" maxOccurs="1" />
999     <xs:element name="openlrBearing" type="D2LogicalModel:AngleInDegrees" minOccurs=
↪ "1" maxOccurs="1" />
1000    <xs:element name="openlrLineAttributesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />

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1001   </xs:sequence>
1002 </xs:complexType>
1003 <xs:complexType name="OpenlrLineLocationReference">
1004   <xs:sequence>
1005     <xs:element name="openlrLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLocationReferencePoint" maxOccurs="unbounded" />
1006     <xs:element name="openlrLastLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
1007     <xs:element name="openlrOffsets" type="D2LogicalModel:OpenlrOffsets" minOccurs=
↪ "0" />
1008     <xs:element name="openlrLineLocationReferenceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1009   </xs:sequence>
1010 </xs:complexType>
1011 <xs:complexType name="OpenlrLocationReferencePoint">
1012   <xs:complexContent>
1013     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1014       <xs:sequence>
1015         <xs:element name="openlrPathAttributes" type=
↪ "D2LogicalModel:OpenlrPathAttributes" />
1016         <xs:element name="openlrLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1017       </xs:sequence>

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1018     </xs:extension>
1019   </xs:complexContent>
1020 </xs:complexType>
1021 <xs:complexType name="OpenlrOffsets">
1022   <xs:sequence>
1023     <xs:element name="openlrPositiveOffset" type=
1024     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1025     <xs:element name="openlrNegativeOffset" type=
1026     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1027     <xs:element name="openlrOffsetsExtension" type="D2LogicalModel:_
1028     ↪ ExtensionType" minOccurs="0" />
1029   </xs:sequence>
1030 </xs:complexType>
1031 <xs:simpleType name="OpenlrOrientationEnum">
1032   <xs:restriction base="xs:string">
1033     <xs:enumeration value="noOrientationOrUnknown" />
1034     <xs:enumeration value="withLineDirection" />
1035     <xs:enumeration value="againstLineDirection" />
1036     <xs:enumeration value="both" />
1037   </xs:restriction>
1038 </xs:simpleType>
1039 <xs:complexType name="OpenlrPathAttributes">
1040   <xs:sequence>
1041     <xs:element name="openlrLowestFRCToNextLRPoint" type=
1042     ↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
1043     <xs:element name="openlrDistanceToNextLRPoint" type=
1044     ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
1045     <xs:element name="openlrPathAttributesExtension" type="D2LogicalModel:_
1046     ↪ ExtensionType" minOccurs="0" />
1047   </xs:sequence>
1048 </xs:complexType>
1049 <xs:complexType name="OpenlrPointAlongLine">
1050   <xs:complexContent>
1051     <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1052       <xs:sequence>
1053         <xs:element name="openlrPointAlongLineExtension" type="D2LogicalModel:_
1054         ↪ ExtensionType" minOccurs="0" />
1055       </xs:sequence>
1056     </xs:extension>
1057   </xs:complexContent>
1058 </xs:complexType>
1059 <xs:complexType name="OpenlrPointLocationReference">
1060   <xs:sequence>
1061     <xs:element name="openlrGeoCoordinate" type="D2LogicalModel:OpenlrGeoCoordinate
1062     ↪ " minOccurs="0" />
1063     <xs:element name="openlrPoiWithAccessPoint" type=
1064     ↪ "D2LogicalModel:OpenlrPoiWithAccessPoint" minOccurs="0" />
1065     <xs:element name="openlrPointAlongLine" type=
1066     ↪ "D2LogicalModel:OpenlrPointAlongLine" minOccurs="0" />
1067     <xs:element name="openlrPointLocationReferenceExtension" type="D2LogicalModel:_
1068     ↪ ExtensionType" minOccurs="0" />
1069   </xs:sequence>
1070 </xs:complexType>
1071 <xs:complexType name="OpenlrPoiWithAccessPoint">
1072   <xs:complexContent>
1073     <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">

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1063     <xs:sequence>
1064     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" /
1065     <xs:element name="openlrPoiWithAccessPointExtension" type="D2LogicalModel:_
1066     <ExtensionType" minOccurs="0" />
1067     </xs:sequence>
1068     </xs:extension>
1069     </xs:complexContent>
1070     </xs:complexType>
1071     <xs:simpleType name="OpenlrSideOfRoadEnum">
1072     <xs:restriction base="xs:string">
1073     <xs:enumeration value="onRoadOrUnknown" />
1074     <xs:enumeration value="right" />
1075     <xs:enumeration value="left" />
1076     <xs:enumeration value="both" />
1077     </xs:restriction>
1078     </xs:simpleType>
1079     <xs:complexType name="PayloadPublication" abstract="true">
1080     <xs:sequence>
1081     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"
1082     <maxOccurs="1" />
1083     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"
1084     <maxOccurs="1" />
1085     <xs:element name="publicationCreator" type=
1086     <"D2LogicalModel:InternationalIdentifier" />
1087     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
1088     <ExtensionType" minOccurs="0" />
1089     </xs:sequence>
1090     <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
1091     </xs:complexType>
1092     <xs:simpleType name="Percentage">
1093     <xs:restriction base="D2LogicalModel:Float" />
1094     </xs:simpleType>
1095     <xs:complexType name="PercentageDistanceAlongLinearElement">
1096     <xs:complexContent>
1097     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
1098     <xs:sequence>
1099     <xs:element name="percentageDistanceAlong" type="D2LogicalModel:Percentage"
1100     <minOccurs="1" maxOccurs="1" />
1101     <xs:element name="percentageDistanceAlongLinearElementExtension" type=
1102     <"D2LogicalModel:_ExtensionType" minOccurs="0" />
1103     </xs:sequence>
1104     </xs:extension>
1105     </xs:complexContent>
1106     </xs:complexType>
1107     <xs:complexType name="Point">
1108     <xs:complexContent>
1109     <xs:extension base="D2LogicalModel:NetworkLocation">
1110     <xs:sequence>
1111     <xs:element name="tpegPointLocation" type="D2LogicalModel:TpegPointLocation
1112     <" minOccurs="0" />
1113     <xs:element name="alertCPoint" type="D2LogicalModel:AlertCPoint" minOccurs=
1114     <"0" />
1115     <xs:element name="pointAlongLinearElement" type=
1116     <"D2LogicalModel:PointAlongLinearElement" minOccurs="0" />
1117     <xs:element name="pointByCoordinates" type=

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1108 < "D2LogicalModel:PointByCoordinates" minOccurs="0" />
      <xs:element name="pointWithinLinearGNElement" type=
1109 < "D2LogicalModel:PointWithinLinearGNElement" minOccurs="0" />
      <xs:element name="pointByMultiCoordinates" type=
1110 < "D2LogicalModel:PointByMultiCoordinates" minOccurs="0" />
      <xs:element name="pointExtension" type="D2LogicalModel:_PointExtensionType"
1111 < minOccurs="0" />
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="PointAlongLinearElement">
  <xs:sequence>
    <xs:element name="administrativeAreaOfPoint" type=
1118 < "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
    <xs:element name="directionBoundAtPoint" type="D2LogicalModel:DirectionEnum"
1119 < minOccurs="0" maxOccurs="1" />
    <xs:element name="directionRelativeAtPoint" type=
1120 < "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
    <xs:element name="heightGradeOfPoint" type="D2LogicalModel:HeightGradeEnum"
1121 < minOccurs="0" maxOccurs="1" />
    <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
    <xs:element name="distanceAlongLinearElement" type=
1122 < "D2LogicalModel:DistanceAlongLinearElement" />
    <xs:element name="pointAlongLinearElementExtension" type="D2LogicalModel:_
1123 < ExtensionType" minOccurs="0" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PointByCoordinates">
  <xs:sequence>
    <xs:element name="bearing" type="D2LogicalModel:NonNegativeInteger" minOccurs="0
1128 < " maxOccurs="1" />
    <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates" />
1129 <xs:element name="pointByCoordinatesExtension" type="D2LogicalModel:_
1130 < ExtensionType" minOccurs="0" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PointByMultiCoordinates">
  <xs:sequence>
    <xs:element name="sjtskPointCoordinates" type=
1135 < "D2LogicalModel:SJTSKPointCoordinates" minOccurs="0" />
    <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
1136 < minOccurs="0" />
    <xs:element name="pointByMultiCoordinatesExtension" type="D2LogicalModel:_
1137 < ExtensionType" minOccurs="0" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PointCoordinates">
  <xs:sequence>
    <xs:element name="latitude" type="D2LogicalModel:Float" minOccurs="1" maxOccurs=
1142 < "1" />
    <xs:element name="longitude" type="D2LogicalModel:Float" minOccurs="1"
1143 < maxOccurs="1" />
    <xs:element name="pointCoordinatesExtension" type="D2LogicalModel:_ExtensionType
1144 < " minOccurs="0" />
  </xs:sequence>
1145

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1146 </xs:complexType>
1147 <xs:complexType name="PointDestination">
1148   <xs:complexContent>
1149     <xs:extension base="D2LogicalModel:Destination">
1150       <xs:sequence>
1151         <xs:element name="point" type="D2LogicalModel:Point" />
1152         <xs:element name="pointDestinationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1153       </xs:sequence>
1154     </xs:extension>
1155   </xs:complexContent>
1156 </xs:complexType>
1157 <xs:complexType name="PointExtended">
1158   <xs:sequence>
1159     <xs:element name="description" type="D2LogicalModel:MultilingualString"
↪ minOccurs="0" maxOccurs="1" />
1160     <xs:element name="junction" type="D2LogicalModel:Junction" minOccurs="0" />
1161   </xs:sequence>
1162 </xs:complexType>
1163 <xs:complexType name="PointWithinLinearGNElement">
1164   <xs:sequence>
1165     <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
↪ "0" maxOccurs="1" />
1166     <xs:element name="directionRelativeOnLinearSection" type=
↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
1167     <xs:element name="networkVersion" type="D2LogicalModel:MultilingualString"
↪ minOccurs="1" maxOccurs="1" />
1168     <xs:element name="distanceAlongLinearElement" type=
↪ "D2LogicalModel:DistanceAlongLinearElement" />
1169     <xs:element name="pointWithinLinearGNElementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1170   </xs:sequence>
1171 </xs:complexType>
1172 <xs:complexType name="Referent">
1173   <xs:sequence>
1174     <xs:element name="referentIdentifier" type="D2LogicalModel:String" minOccurs="1"
↪ maxOccurs="1" />
1175     <xs:element name="referentName" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
1176     <xs:element name="referentType" type="D2LogicalModel:ReferentTypeEnum"
↪ minOccurs="1" maxOccurs="1" />
1177     <xs:element name="referentDescription" type="D2LogicalModel:MultilingualString"
↪ minOccurs="0" maxOccurs="1" />
1178     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
↪ minOccurs="0" />
1179     <xs:element name="referentExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
1180   </xs:sequence>
1181 </xs:complexType>
1182 <xs:simpleType name="ReferentTypeEnum">
1183   <xs:restriction base="xs:string">
1184     <xs:enumeration value="boundary" />
1185     <xs:enumeration value="intersection" />
1186     <xs:enumeration value="referenceMarker" />
1187     <xs:enumeration value="landmark" />
1188     <xs:enumeration value="roadNode" />

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1189     </xs:restriction>
1190 </xs:simpleType>
1191 <xs:complexType name="Road">
1192   <xs:sequence>
1193     <xs:element name="nameOfRoad" type="D2LogicalModel:MultilingualString"
1194     ↪ minOccurs="0" maxOccurs="1" />
1195     <xs:element name="roadIdentifier" type="D2LogicalModel:MultilingualString"
1196     ↪ minOccurs="0" maxOccurs="1" />
1197     <xs:element name="typeOfRoad" type="D2LogicalModel:RoadTypeEnum" minOccurs="0"
1198     ↪ maxOccurs="1" />
1199     <xs:element name="roadDestination" type="D2LogicalModel:MultilingualString"
1200     ↪ minOccurs="0" maxOccurs="unbounded" />
1201     <xs:element name="roadOrigination" type="D2LogicalModel:MultilingualString"
1202     ↪ minOccurs="0" maxOccurs="unbounded" />
1203     <xs:element name="distanceToThisRoad" type=
1204     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1205     <xs:element name="roadExtension" type="D2LogicalModel:_ExtensionType" minOccurs=
1206     ↪ "0" />
1207   </xs:sequence>
1208 </xs:complexType>
1209 <xs:complexType name="RoadNode">
1210   <xs:complexContent>
1211     <xs:extension base="D2LogicalModel:Road">
1212       <xs:sequence>
1213         <xs:element name="junctionName" type="D2LogicalModel:MultilingualString"
1214         ↪ minOccurs="1" maxOccurs="1" />
1215         <xs:element name="roadNodeExtension" type="D2LogicalModel:_ExtensionType"
1216         ↪ minOccurs="0" />
1217       </xs:sequence>
1218     </xs:extension>
1219   </xs:complexContent>
1220 </xs:complexType>
1221 <xs:simpleType name="RoadTypeEnum">
1222   <xs:restriction base="xs:string">
1223     <xs:enumeration value="motorway" />
1224     <xs:enumeration value="trunkRoad" />
1225     <xs:enumeration value="mainRoad" />
1226     <xs:enumeration value="other" />
1227   </xs:restriction>
1228 </xs:simpleType>
1229 <xs:simpleType name="Seconds">
1230   <xs:restriction base="D2LogicalModel:Float" />
1231 </xs:simpleType>
1232 <xs:complexType name="SJTSKPointCoordinates">
1233   <xs:sequence>
1234     <xs:element name="sjtskX" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
1235     ↪ />
1236     <xs:element name="sjtskY" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
1237     ↪ />
1238     <xs:element name="sjtskPointCoordinatesExtension" type="D2LogicalModel:_
1239     ↪ ExtensionType" minOccurs="0" />
1240   </xs:sequence>
1241 </xs:complexType>
1242 <xs:simpleType name="String">
1243   <xs:restriction base="xs:string">
1244     <xs:maxLength value="1024" />

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1233     </xs:restriction>
1234 </xs:simpleType>
1235 <xs:complexType name="SupplementaryPositionalDescription">
1236   <xs:sequence>
1237     <xs:element name="locationDescriptor" type=
1238 ↪ "D2LogicalModel:LocationDescriptorEnum" minOccurs="0" maxOccurs="unbounded" />
1239     <xs:element name="sequentialRampNumber" type="D2LogicalModel:NonNegativeInteger"
1240 ↪ " minOccurs="0" maxOccurs="1" />
1241     <xs:element name="affectedCarriagewayAndLanes" type=
1242 ↪ "D2LogicalModel:AffectedCarriagewayAndLanes" minOccurs="0" maxOccurs="unbounded" />
1243     <xs:element name="supplementaryPositionalDescriptionExtension" type=
1244 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1245   </xs:sequence>
1246   <xs:attribute name="locationPrecision" type=
1247 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" use="optional" />
1248 </xs:complexType>
1249 <xs:simpleType name="Tonnes">
1250   <xs:restriction base="D2LogicalModel:Float" />
1251 </xs:simpleType>
1252 <xs:complexType name="TpegDescriptor" abstract="true">
1253   <xs:sequence>
1254     <xs:element name="descriptor" type="D2LogicalModel:MultilingualString"
1255 ↪ minOccurs="1" maxOccurs="1" />
1256     <xs:element name="tppegDescriptorExtension" type="D2LogicalModel:_ExtensionType"
1257 ↪ minOccurs="0" />
1258   </xs:sequence>
1259 </xs:complexType>
1260 <xs:complexType name="TpegFramedPoint">
1261   <xs:complexContent>
1262     <xs:extension base="D2LogicalModel:TpegPointLocation">
1263       <xs:sequence>
1264         <xs:element name="tppegFramedPointLocationType" type=
1265 ↪ "D2LogicalModel:TpegLoc01FramedPointLocationSubtypeEnum" minOccurs="1" maxOccurs="1"
1266 ↪ />
1267         <xs:element name="framedPoint" type="D2LogicalModel:TpegNonJunctionPoint" />
1268         <xs:element name="to" type="D2LogicalModel:TpegPoint" />
1269         <xs:element name="from" type="D2LogicalModel:TpegPoint" />
1270         <xs:element name="tppegFramedPointExtension" type="D2LogicalModel:_
1271 ↪ ExtensionType" minOccurs="0" />
1272       </xs:sequence>
1273     </xs:extension>
1274   </xs:complexContent>
1275 </xs:complexType>
1276 <xs:complexType name="TpegIlcPointDescriptor">
1277   <xs:complexContent>
1278     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
1279       <xs:sequence>
1280         <xs:element name="tppegIlcPointDescriptorType" type=
1281 ↪ "D2LogicalModel:TpegLoc03IlcPointDescriptorSubtypeEnum" minOccurs="1" maxOccurs="1"
1282 ↪ />
1283         <xs:element name="tppegIlcPointDescriptorExtension" type="D2LogicalModel:_
1284 ↪ ExtensionType" minOccurs="0" />
1285       </xs:sequence>
1286     </xs:extension>
1287   </xs:complexContent>
1288 </xs:complexType>

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1276 <xs:complexType name="TpegJunction">
1277   <xs:complexContent>
1278     <xs:extension base="D2LogicalModel:TpegPoint">
1279       <xs:sequence>
1280         <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates" /
1281         <xs:element name="name" type="D2LogicalModel:TpegJunctionPointDescriptor"
1282         <xs:element name="ilc" type="D2LogicalModel:TpegIlcPointDescriptor"
1283         <xs:element name="otherName" type="D2LogicalModel:TpegOtherPointDescriptor"
1284         <xs:element name="tpegJunctionExtension" type="D2LogicalModel:_ExtensionType
1285       </xs:sequence>
1286     </xs:extension>
1287   </xs:complexContent>
1288 </xs:complexType>
1289 <xs:complexType name="TpegJunctionPointDescriptor">
1290   <xs:complexContent>
1291     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
1292       <xs:sequence>
1293         <xs:element name="tpegJunctionPointDescriptorType" type=
1294         <xs:element name="tpegJunctionPointDescriptorExtension" type=
1295       </xs:sequence>
1296     </xs:extension>
1297   </xs:complexContent>
1298 </xs:complexType>
1299 <xs:complexType name="TpegLinearLocation">
1300   <xs:sequence>
1301     <xs:element name="tpegDirection" type="D2LogicalModel:DirectionEnum" minOccurs=
1302     <xs:element name="tpegLinearLocationType" type=
1303     <xs:element name="to" type="D2LogicalModel:TpegPoint" />
1304     <xs:element name="from" type="D2LogicalModel:TpegPoint" />
1305     <xs:element name="tpegLinearLocationExtension" type="D2LogicalModel:_
1306   </xs:sequence>
1307 </xs:complexType>
1308 <xs:simpleType name="TpegLoc01FramedPointLocationSubtypeEnum">
1309   <xs:restriction base="xs:string">
1310     <xs:enumeration value="framedPoint" />
1311   </xs:restriction>
1312 </xs:simpleType>
1313 <xs:simpleType name="TpegLoc01LinearLocationSubtypeEnum">
1314   <xs:restriction base="xs:string">
1315     <xs:enumeration value="segment" />
1316   </xs:restriction>
1317 </xs:simpleType>
1318 <xs:simpleType name="TpegLoc01SimplePointLocationSubtypeEnum">
1319   <xs:restriction base="xs:string">
1320     <xs:enumeration value="intersection" />

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1321     <xs:enumeration value="nonLinkedPoint" />
1322   </xs:restriction>
1323 </xs:simpleType>
1324 <xs:simpleType name="TpegLoc03IlcPointDescriptorSubtypeEnum">
1325   <xs:restriction base="xs:string">
1326     <xs:enumeration value="tpegIlcName1" />
1327     <xs:enumeration value="tpegIlcName2" />
1328     <xs:enumeration value="tpegIlcName3" />
1329   </xs:restriction>
1330 </xs:simpleType>
1331 <xs:simpleType name="TpegLoc03JunctionPointDescriptorSubtypeEnum">
1332   <xs:restriction base="xs:string">
1333     <xs:enumeration value="junctionName" />
1334   </xs:restriction>
1335 </xs:simpleType>
1336 <xs:simpleType name="TpegLoc03OtherPointDescriptorSubtypeEnum">
1337   <xs:restriction base="xs:string">
1338     <xs:enumeration value="administrativeAreaName" />
1339     <xs:enumeration value="administrativeReferenceName" />
1340     <xs:enumeration value="airportName" />
1341     <xs:enumeration value="areaName" />
1342     <xs:enumeration value="buildingName" />
1343     <xs:enumeration value="busStopIdentifier" />
1344     <xs:enumeration value="busStopName" />
1345     <xs:enumeration value="canalName" />
1346     <xs:enumeration value="countyName" />
1347     <xs:enumeration value="ferryPortName" />
1348     <xs:enumeration value="intersectionName" />
1349     <xs:enumeration value="lakeName" />
1350     <xs:enumeration value="linkName" />
1351     <xs:enumeration value="localLinkName" />
1352     <xs:enumeration value="metroStationName" />
1353     <xs:enumeration value="nationName" />
1354     <xs:enumeration value="nonLinkedPointName" />
1355     <xs:enumeration value="parkingFacilityName" />
1356     <xs:enumeration value="pointName" />
1357     <xs:enumeration value="pointOfInterestName" />
1358     <xs:enumeration value="railwayStation" />
1359     <xs:enumeration value="regionName" />
1360     <xs:enumeration value="riverName" />
1361     <xs:enumeration value="seaName" />
1362     <xs:enumeration value="serviceAreaName" />
1363     <xs:enumeration value="tidalRiverName" />
1364     <xs:enumeration value="townName" />
1365     <xs:enumeration value="other" />
1366   </xs:restriction>
1367 </xs:simpleType>
1368 <xs:complexType name="TpegNonJunctionPoint">
1369   <xs:complexContent>
1370     <xs:extension base="D2LogicalModel:TpegPoint">
1371       <xs:sequence>
1372         <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates" /
1373         <xs:element name="name" type="D2LogicalModel:TpegOtherPointDescriptor"
1374         <xs:element name="tpgNonJunctionPointExtension" type="D2LogicalModel:_

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1375     <xs:extensionType" minOccurs="0" />
1376     </xs:sequence>
1377   </xs:extension>
1378 </xs:complexContent>
1379 </xs:complexType>
1380 <xs:complexType name="TpegOtherPointDescriptor">
1381   <xs:complexContent>
1382     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
1383       <xs:sequence>
1384         <xs:element name="tpegOtherPointDescriptorType" type=
1385         <xs:extensionType" minOccurs="0" />
1386         </xs:sequence>
1387       </xs:extension>
1388     </xs:complexContent>
1389   </xs:complexType>
1390   <xs:complexType name="TpegPoint" abstract="true">
1391     <xs:sequence>
1392       <xs:element name="tpegPointExtension" type="D2LogicalModel:_
1393       <minOccurs="0" />
1394       </xs:sequence>
1395     </xs:complexType>
1396     <xs:complexType name="TpegPointDescriptor" abstract="true">
1397       <xs:complexContent>
1398         <xs:extension base="D2LogicalModel:TpegDescriptor">
1399           <xs:sequence>
1400             <xs:element name="tpegPointDescriptorExtension" type="D2LogicalModel:_
1401             <xs:extensionType" minOccurs="0" />
1402             </xs:sequence>
1403           </xs:extension>
1404         </xs:complexContent>
1405       </xs:complexType>
1406       <xs:complexType name="TpegPointLocation" abstract="true">
1407         <xs:sequence>
1408           <xs:element name="tpegDirection" type="D2LogicalModel:DirectionEnum" minOccurs=
1409           <xs:extensionType" minOccurs="0" />
1410           </xs:sequence>
1411         </xs:complexType>
1412         <xs:complexType name="TpegSimplePoint">
1413           <xs:complexContent>
1414             <xs:extension base="D2LogicalModel:TpegPointLocation">
1415               <xs:sequence>
1416                 <xs:element name="tpegSimplePointLocationType" type=
1417                 <xs:extensionType" minOccurs="0" />
1418                 </xs:sequence>
1419               </xs:extension>
1420             </xs:complexContent>
1421           </xs:complexType>

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1420 <xs:simpleType name="UrgencyEnum">
1421   <xs:restriction base="xs:string">
1422     <xs:enumeration value="extremelyUrgent" />
1423     <xs:enumeration value="urgent" />
1424     <xs:enumeration value="normalUrgency" />
1425   </xs:restriction>
1426 </xs:simpleType>
1427 <xs:complexType name="VehicleCharacteristics">
1428   <xs:sequence>
1429     <xs:element name="fuelType" type="D2LogicalModel:FuelTypeEnum" minOccurs="0"
↪ maxOccurs="1" />
1430     <xs:element name="loadType" type="D2LogicalModel:LoadTypeEnum" minOccurs="0"
↪ maxOccurs="1" />
1431     <xs:element name="vehicleEquipment" type="D2LogicalModel:VehicleEquipmentEnum"
↪ minOccurs="0" maxOccurs="1" />
1432     <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
↪ "0" maxOccurs="unbounded" />
1433     <xs:element name="vehicleUsage" type="D2LogicalModel:VehicleUsageEnum"
↪ minOccurs="0" maxOccurs="1" />
1434     <xs:element name="grossWeightCharacteristic" type=
↪ "D2LogicalModel:GrossWeightCharacteristic" minOccurs="0" maxOccurs="2" />
1435     <xs:element name="heightCharacteristic" type=
↪ "D2LogicalModel:HeightCharacteristic" minOccurs="0" maxOccurs="2" />
1436     <xs:element name="lengthCharacteristic" type=
↪ "D2LogicalModel:LengthCharacteristic" minOccurs="0" maxOccurs="2" />
1437     <xs:element name="widthCharacteristic" type="D2LogicalModel:WidthCharacteristic
↪ " minOccurs="0" maxOccurs="2" />
1438     <xs:element name="heaviestAxleWeightCharacteristic" type=
↪ "D2LogicalModel:HeaviestAxleWeightCharacteristic" minOccurs="0" maxOccurs="2" />
1439     <xs:element name="numberOfAxlesCharacteristic" type=
↪ "D2LogicalModel:NumberOfAxlesCharacteristic" minOccurs="0" maxOccurs="2" />
1440     <xs:element name="vehicleCharacteristicsExtension" type="D2LogicalModel:_
↪ VehicleCharacteristicsExtensionType" minOccurs="0" />
1441   </xs:sequence>
1442 </xs:complexType>
1443 <xs:complexType name="VehicleCharacteristicsExtended">
1444   <xs:sequence>
1445     <xs:element name="emissionClassification" type="D2LogicalModel:String"
↪ minOccurs="0" maxOccurs="unbounded" />
1446     <xs:element name="operationFreeOfEmission" type="D2LogicalModel:Boolean"
↪ minOccurs="0" maxOccurs="1" />
1447     <xs:element name="loadType2" type="D2LogicalModel:LoadType2Enum" minOccurs="0"
↪ maxOccurs="1" />
1448     <xs:element name="vehicleType2" type="D2LogicalModel:VehicleType2Enum"
↪ minOccurs="0" maxOccurs="1" />
1449     <xs:element name="fuelType2" type="D2LogicalModel:FuelType2Enum" minOccurs="0"
↪ maxOccurs="1" />
1450     <xs:element name="vehicleUsage2" type="D2LogicalModel:VehicleUsage2Enum"
↪ minOccurs="0" maxOccurs="1" />
1451   </xs:sequence>
1452 </xs:complexType>
1453 <xs:complexType name="VehicleCountValue">
1454   <xs:complexContent>
1455     <xs:extension base="D2LogicalModel:DataValue">
1456       <xs:sequence>
1457         <xs:element name="vehicleCount" type="D2LogicalModel:NonNegativeInteger"

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1458 <minOccurs="1" maxOccurs="1" />
      <xs:element name="vehicleCountValueExtension" type="D2LogicalModel:_
1459 <ExtensionType" minOccurs="0" />
1460   </xs:sequence>
1461 </xs:extension>
1462 </xs:complexContent>
1463 </xs:complexType>
1464 <xs:simpleType name="VehicleEquipmentEnum">
1465   <xs:restriction base="xs:string">
1466     <xs:enumeration value="notUsingSnowChains" />
1467     <xs:enumeration value="notUsingSnowChainsOrTyres" />
1468     <xs:enumeration value="snowChainsInUse" />
1469     <xs:enumeration value="snowTyresInUse" />
1470     <xs:enumeration value="snowChainsOrTyresInUse" />
1471     <xs:enumeration value="withoutSnowTyresOrChainsOnBoard" />
1472   </xs:restriction>
1473 </xs:simpleType>
1474 <xs:simpleType name="VehicleType2Enum">
1475   <xs:restriction base="xs:string">
1476     <xs:enumeration value="motorhome" />
1477     <xs:enumeration value="lightGoodsVehicle" />
1478     <xs:enumeration value="heavyGoodsVehicle" />
1479     <xs:enumeration value="minibus" />
1480     <xs:enumeration value="smallCar" />
1481     <xs:enumeration value="largeCar" />
1482     <xs:enumeration value="lightGoodsVehicleWithTrailer" />
1483     <xs:enumeration value="heavyGoodsVehicleWithTrailer" />
1484     <xs:enumeration value="heavyHaulageVehicle" />
1485     <xs:enumeration value="passengerCar" />
1486     <xs:enumeration value="unknown" />
1487   </xs:restriction>
1488 </xs:simpleType>
1489 <xs:simpleType name="VehicleTypeEnum">
1490   <xs:restriction base="xs:string">
1491     <xs:enumeration value="agriculturalVehicle" />
1492     <xs:enumeration value="anyVehicle" />
1493     <xs:enumeration value="articulatedVehicle" />
1494     <xs:enumeration value="bicycle" />
1495     <xs:enumeration value="bus" />
1496     <xs:enumeration value="car" />
1497     <xs:enumeration value="caravan" />
1498     <xs:enumeration value="carOrLightVehicle" />
1499     <xs:enumeration value="carWithCaravan" />
1500     <xs:enumeration value="carWithTrailer" />
1501     <xs:enumeration value="constructionOrMaintenanceVehicle" />
1502     <xs:enumeration value="fourWheelDrive" />
1503     <xs:enumeration value="highSidedVehicle" />
1504     <xs:enumeration value="lorry" />
1505     <xs:enumeration value="moped" />
1506     <xs:enumeration value="motorcycle" />
1507     <xs:enumeration value="motorcycleWithSideCar" />
1508     <xs:enumeration value="motorscooter" />
1509     <xs:enumeration value="tanker" />
1510     <xs:enumeration value="threeWheeledVehicle" />
1511     <xs:enumeration value="trailer" />
      <xs:enumeration value="tram" />

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1512     <xs:enumeration value="twoWheeledVehicle" />
1513     <xs:enumeration value="van" />
1514     <xs:enumeration value="vehicleWithCatalyticConverter" />
1515     <xs:enumeration value="vehicleWithoutCatalyticConverter" />
1516     <xs:enumeration value="vehicleWithCaravan" />
1517     <xs:enumeration value="vehicleWithTrailer" />
1518     <xs:enumeration value="withEvenNumberedRegistrationPlates" />
1519     <xs:enumeration value="withOddNumberedRegistrationPlates" />
1520     <xs:enumeration value="other" />
1521 </xs:restriction>
1522 </xs:simpleType>
1523 <xs:simpleType name="VehicleUsage2Enum">
1524     <xs:restriction base="xs:string">
1525         <xs:enumeration value="cityLogistics" />
1526         <xs:enumeration value="carSharing" />
1527     </xs:restriction>
1528 </xs:simpleType>
1529 <xs:simpleType name="VehicleUsageEnum">
1530     <xs:restriction base="xs:string">
1531         <xs:enumeration value="agricultural" />
1532         <xs:enumeration value="commercial" />
1533         <xs:enumeration value="emergencyServices" />
1534         <xs:enumeration value="military" />
1535         <xs:enumeration value="nonCommercial" />
1536         <xs:enumeration value="patrol" />
1537         <xs:enumeration value="recoveryServices" />
1538         <xs:enumeration value="roadMaintenanceOrConstruction" />
1539         <xs:enumeration value="roadOperator" />
1540         <xs:enumeration value="taxi" />
1541     </xs:restriction>
1542 </xs:simpleType>
1543 <xs:complexType name="VersionedReference">
1544     <xs:attribute name="id" type="xs:string" use="required" />
1545     <xs:attribute name="version" type="xs:string" use="required" />
1546 </xs:complexType>
1547 <xs:complexType name="VmsFault">
1548     <xs:complexContent>
1549         <xs:extension base="D2LogicalModel:Fault">
1550             <xs:sequence>
1551                 <xs:element name="vmsFault" type="D2LogicalModel:VmsFaultEnum" minOccurs="1
1552                 ↪" maxOccurs="1" />
1553                 <xs:element name="vmsFaultExtension" type="D2LogicalModel:_ExtensionType"
1554                 ↪minOccurs="0" />
1555             </xs:sequence>
1556         </xs:extension>
1557     </xs:complexContent>
1558 </xs:complexType>
1559 <xs:simpleType name="VmsFaultEnum">
1560     <xs:restriction base="xs:string">
1561         <xs:enumeration value="communicationsFailure" />
1562         <xs:enumeration value="incorrectMessageDisplayed" />
1563         <xs:enumeration value="incorrectPictogramDisplayed" />
1564         <xs:enumeration value="outOfService" />
1565         <xs:enumeration value="powerFailure" />
1566         <xs:enumeration value="unableToClearDown" />
1567         <xs:enumeration value="unknown" />

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1566     <xs:enumeration value="other" />
1567   </xs:restriction>
1568 </xs:simpleType>
1569 <xs:complexType name="VmsUnitFault">
1570   <xs:complexContent>
1571     <xs:extension base="D2LogicalModel:Fault">
1572       <xs:sequence>
1573         <xs:element name="vmsUnitFault" type="D2LogicalModel:VmsFaultEnum"
1574           ↪ minOccurs="1" maxOccurs="1" />
1575         <xs:element name="vmsUnitFaultExtension" type="D2LogicalModel:_ExtensionType"
1576           ↪ minOccurs="0" />
1577       </xs:sequence>
1578     </xs:extension>
1579   </xs:complexContent>
1580 </xs:complexType>
1581 <xs:complexType name="WidthCharacteristic">
1582   <xs:sequence>
1583     <xs:element name="comparisonOperator" type=
1584       ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1585     <xs:element name="vehicleWidth" type="D2LogicalModel:MetresAsFloat" minOccurs="1"
1586       ↪ maxOccurs="1" />
1587     <xs:element name="widthCharacteristicExtension" type="D2LogicalModel:_
1588       ↪ ExtensionType" minOccurs="0" />
1589   </xs:sequence>
1590 </xs:complexType>
1591 </xs:schema>

```

CHANGELOG

version 1.0.0

- Initial format and documentation

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- [d2supp] DATEX II Current version/Supporting, <https://docs.datex2.eu/>.

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C level extension, [3](#)

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